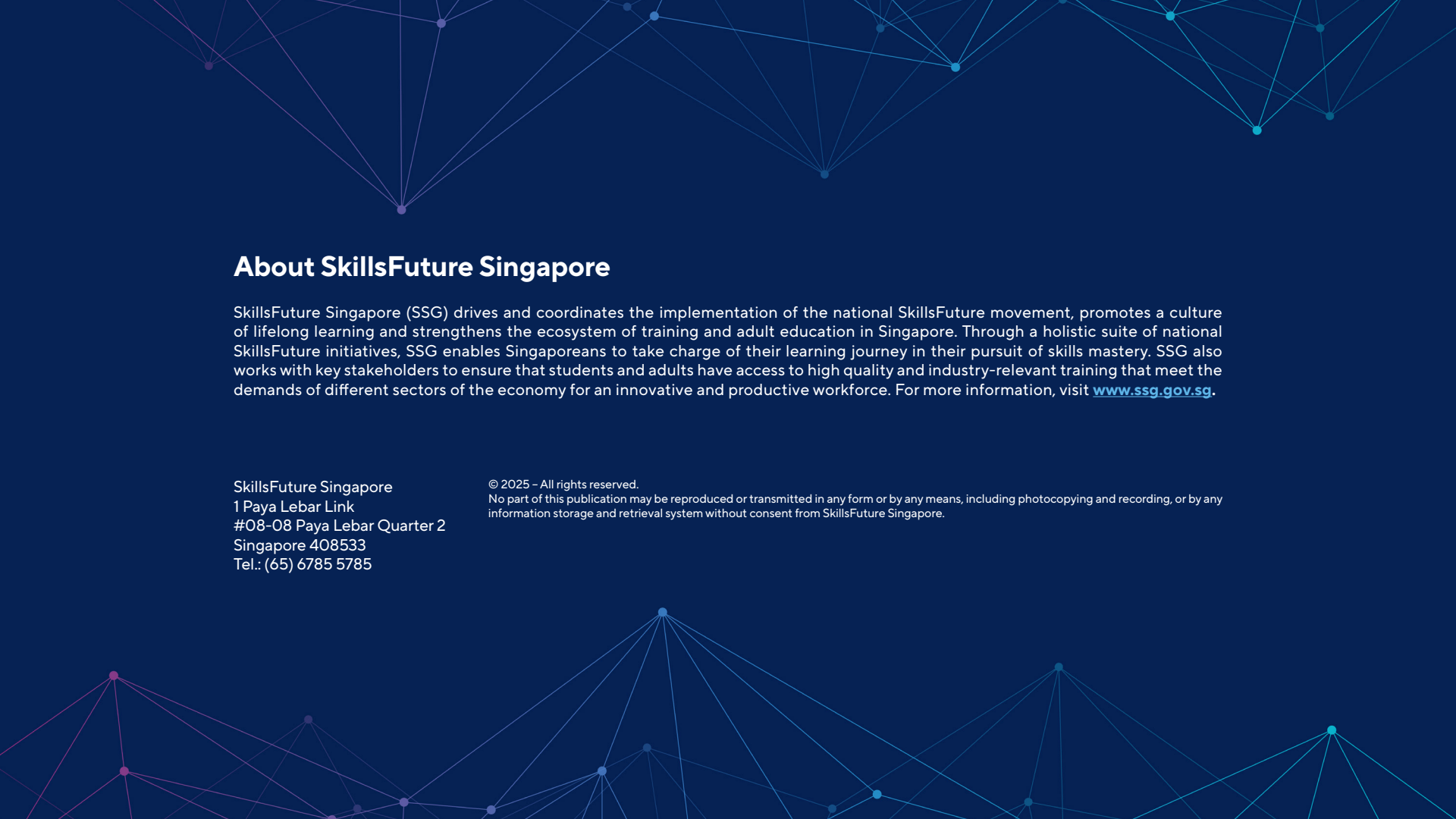


SKILLS DEMAND FOR THE FUTURE ECONOMY REPORT 2025



SKILLS *future* SG



About SkillsFuture Singapore

SkillsFuture Singapore (SSG) drives and coordinates the implementation of the national SkillsFuture movement, promotes a culture of lifelong learning and strengthens the ecosystem of training and adult education in Singapore. Through a holistic suite of national SkillsFuture initiatives, SSG enables Singaporeans to take charge of their learning journey in their pursuit of skills mastery. SSG also works with key stakeholders to ensure that students and adults have access to high quality and industry-relevant training that meet the demands of different sectors of the economy for an innovative and productive workforce. For more information, visit www.ssg.gov.sg.

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What Is In Store For You?

Dr Gog Soon Joo

Chief Skills Officer
SkillsFuture Singapore

The 2025 edition of the Skills Demand for the Future Economy (SDFE) report draws inspiration from our numerous partners and users of the jobs-skills data sets and dashboards released in 2023.

We are heartened by the continued enthusiasm of our skills ecosystem partners utilising these resources to develop targeted insights and solutions (see [“Exemplary Users of SkillsFuture Singapore’s Jobs-Skills Resources”](#) for innovative usage examples). To better serve the skills ecosystem, we have developed a dedicated [Jobs-Skills Portal](#) featuring datasets, dashboards, and resources, enabling users to pull relevant data and insights for their own decision-making. Please register [here](#) to receive regular updates on jobs-skills data and insights.

This edition includes several enhancements and new dashboards to better meet users’ needs:

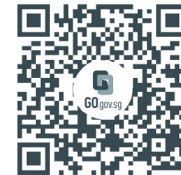
- Linking skills and functions at work: We have revamped the skill clusters to enable readers to connect in-demand skills to business functions, providing a sharper understanding of skills requirements for specific job

roles. In light of technological advancements, we have crowdsourced from global experts to compile a comprehensive list of Applications and Tools (Apps & Tools) used in respective work functions. [Chapter 1](#) offers insights into how skills demands for different job roles have evolved over the past five years and which Apps & Tools have become more prevalent. Readers are encouraged to explore the full list of occupations, corresponding skills requirement changes, and Apps & Tools usage on the [Job Requirements Dashboard](#).

- Skills Matrix: We have also developed a skills matrix to monitor trends in the demand of skills and transferability across job roles. [Chapter 2](#) provides the skills movements over the last three years and skills forecast for 2025 in the Care, Digital, and Green economies. Access the [Priority Skills Dashboard](#) via the portal.
- Enhanced identification of pathways leading to in-demand jobs: In partnership with Workforce Singapore, we have further improved our analysis of identifying potential pathways to support individuals’ career transitions ([Chapter 4](#)). An online [Skills and Job Mobility Dashboard](#) allows readers to explore potential career moves based on criteria such as wage, demand, growth, skills similarity, and historical transitions. The dashboard highlights 146 job roles with good potential for career mobility.

This year’s report also includes a special feature on Singapore’s Creative Industries ([Chapter 3](#)), which have been at the forefront of embracing and integrating technologies in their businesses. Readers can explore the changing skills demands of creative job roles and understand the AI-related Apps & Tools commonly used by job roles in both creative and non-creative sectors via the [Job Requirements Dashboard](#).

Over the past four years, the SDFE editorial team has collaborated with industry professionals, scientists, and researchers from various sectors and professional practices to illuminate Singapore’s skills demand, signpost skills areas to invest in, and support upskilling and reskilling efforts. One thing is abundantly clear: we need all hands on deck in the skills development journey. SkillsFuture Singapore looks forward to your continued partnership in this endeavour.



Click here to access
the Jobs-Skills Portal

Endorsement from Partners



The 2025 SDFE report provides an essential, updated insight into Singapore's evolving job market, focusing on the Care, Digital, and Green economies. Career practitioners will greatly benefit, as they can now guide their clients based on the latest research information. ”

- Timothy Hsi, EdD
President, Career Development Association
of Singapore



The 2025 SDFE report revolutionizes workforce intelligence through its sophisticated mapping of skills, technology, and career pathways. This innovative blueprint empowers businesses to anticipate talent needs while enabling individuals to chart meaningful career transitions that address critical industry demands. Its groundbreaking analysis of emerging technologies and strategic industries creates a dynamic ecosystem where business innovation, personal aspirations, and economic priorities converge. This sophisticated approach to skills intelligence stands as a cornerstone for Singapore's future, while its multilayered methodology ensures our workforce and businesses remain agile, competitive, and future-ready in the global digital economy. ”

- Yustina Saleh, Ph.D.
Managing Director of Innovation Solutions, The Burning Glass Institute

Endorsement from Partners



This report is an important resource for our local creative freelancers to thrive in an innovation-driven economy. I appreciate the report's specific data points that are useful references, for freelancers and micro-enterprises who must navigate a steep upward trajectory of technological change while competing in a globalised and demanding business environment. »

- Jean See
Director, National Trades Union Congress
Freelancers and Self-employed Unit



Insights on how skills requirements are changing for different job roles, and which skills are highly transferable and emerging as priority future skills, are critical for the readiness and resilience of current and future workers in the rapidly changing world of work. The report brings in critical insights on how skills can assist job transition and career development. »

- Akiko Sakamoto, Ph.D.
Senior Regional Skills and Employability
Specialist, International Labour Organisation



The Skills Demand for the Future Economy report's inclusion of relevant applications and tools further aids HR professionals look for candidates possessing the right training to be productive from the get-go. Individuals and HR units can use the Skills and Job Mobility Dashboard to help plan out pathways for current employers to make the most of their training budgets to ensure employee development and job satisfaction. »

- Kan Min-Yen, Ph.D.
Vice Dean (Undergraduate Office),
Associate Professor, NUS School of
Computing, National University of Singapore

CHAPTER 1

Changes in Requirements of Job Roles

Using five years of job posting data from 2019 to 2023, this chapter demonstrates how skill requirements of job roles, as well as use of digital applications and tools (referred to in this report henceforth as **Apps & Tools**), at the workplace have changed over the years. This reflects the changing job requirements for employees.

In this edition, the number of **skill clusters** was simplified from eleven to six to relate more closely with broad business functions and **Critical Core Skills**. The refinement of clusters allows for simpler interpretation of skills demand of jobs. With the proliferation of Apps & Tools across job roles in the respective work functions¹, a comprehensive list of Apps & Tools* is also curated.

The [Job Requirements Dashboard](#) is available at the [Jobs-Skills Portal](#), for readers to explore additional insights.

Click [here](#) to watch videos showcasing our partners' perspectives.

* The Apps & Tools list is constantly being improved and refined.

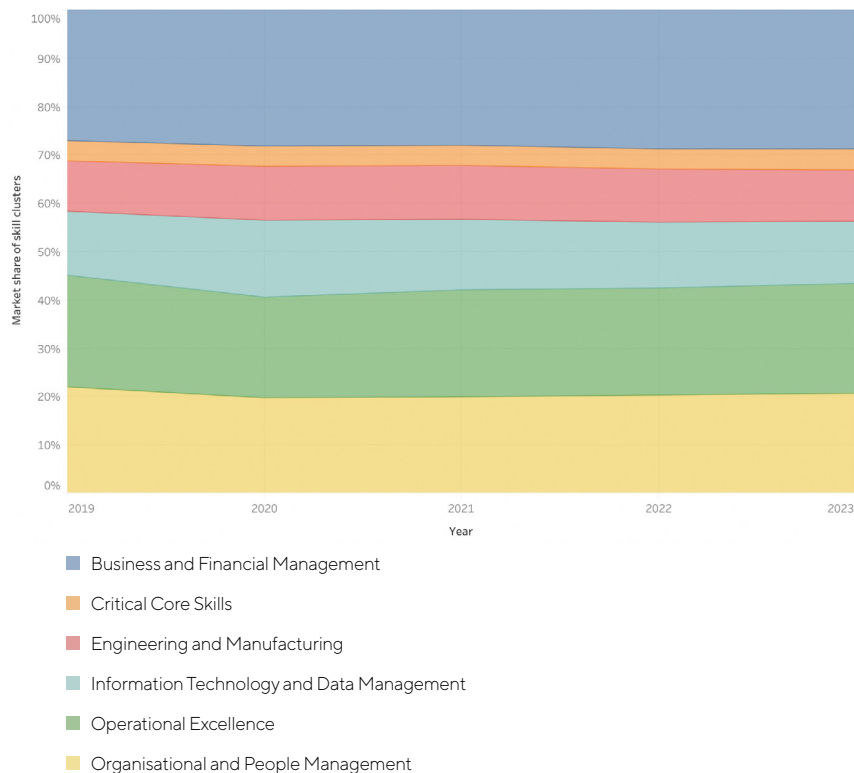
Skill requirements for Singapore as a whole remained relatively stable.

The three largest skill clusters demanded across job roles are:

- Business and Financial Management
- Operational Excellence
- Organisational and People Management

To explore the changes in job requirement of specific job roles, readers are encouraged to use the interactive chart on the [Jobs-Skills Portal](#).

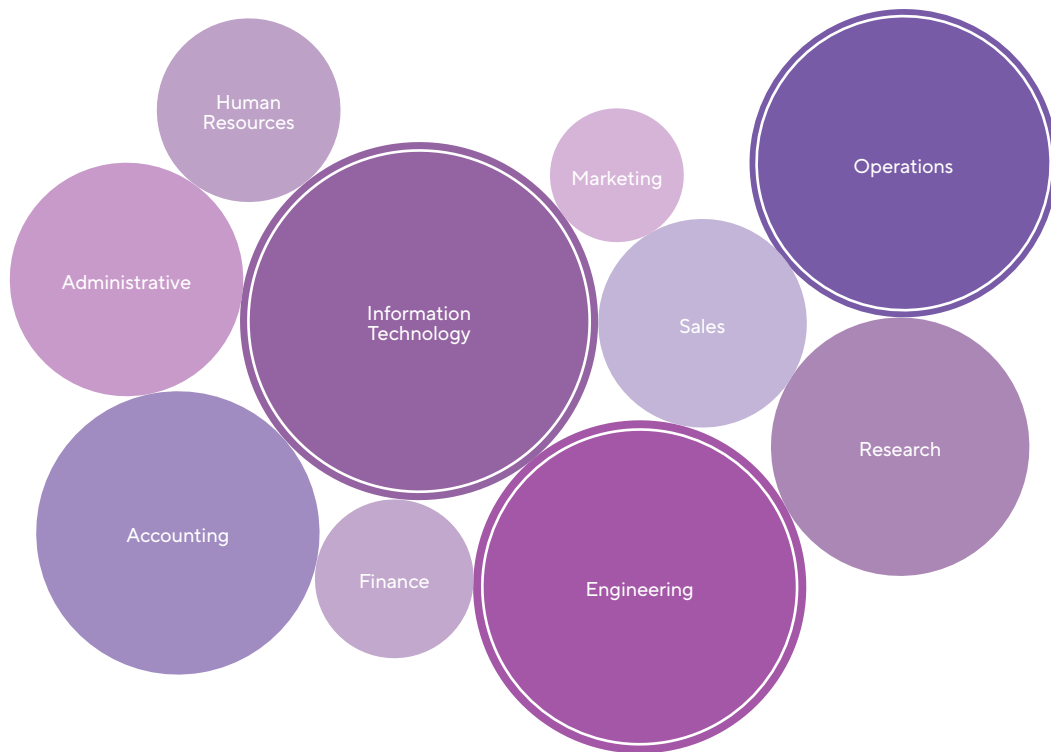
Skill Clusters of The Singapore Economy



Apps & Tools were most used in work functions¹ related to Information Technology (IT), Engineering, and Operations.

The top three work functions that use Apps & Tools are IT (25%), followed by Engineering (13%), and Operations (8%), which make up 46% of all Apps & Tools required. Examples of Apps & Tools most required in these work functions are *Python*, *AutoCAD*, and *SAP Warehouse*, respectively.

Demand of Apps & Tools by Work Function in 2023



¹ Work functions are the core activities or tasks that an individual performs as part of their job role. These work functions are developed with reference to common business functions found in most organisations

The top 20 Apps & Tools experienced changes in relative demand between 2019 and 2023.

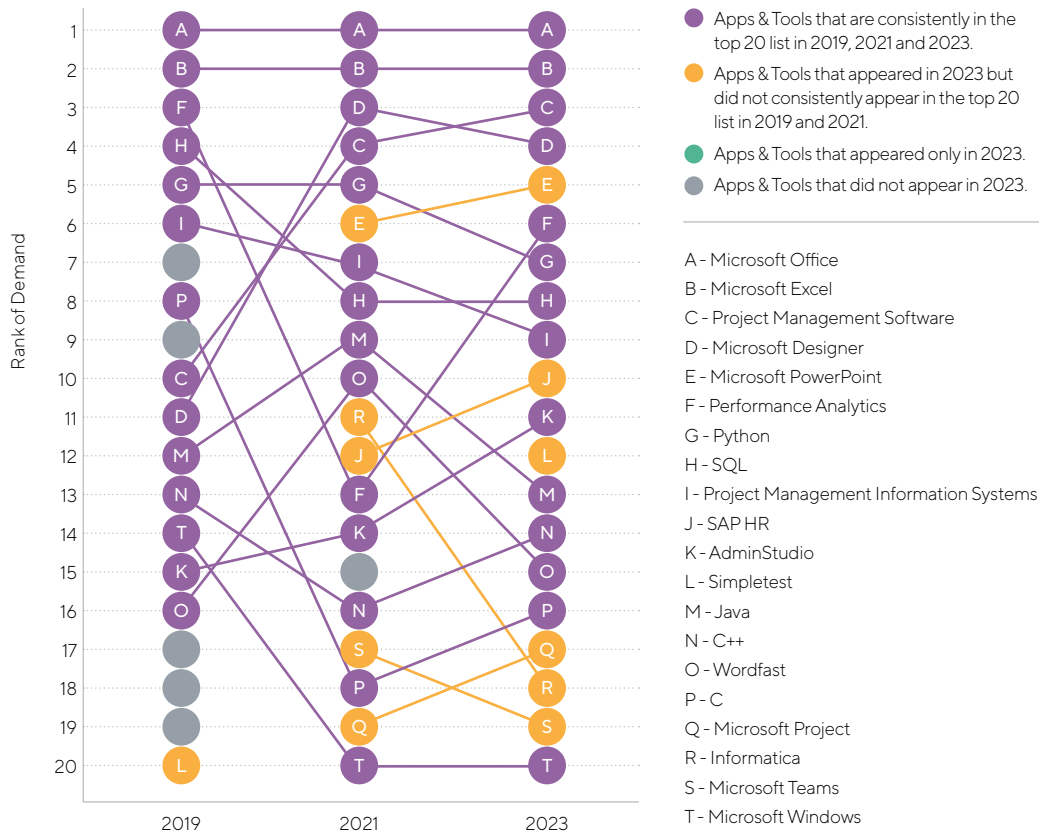
Microsoft Teams (moved from 31st to 19th) and *Microsoft Project* (26th to 17th) rose in rank over the last five years, indicating that they were becoming more frequently utilised.

Notably, *Wordfast*, a translation software, saw a sharp decline in 2020 before recovering its position in 2023 (ranked 15th).

In 2023, the top 20 Apps & Tools comprised mainly of productivity tools, programming languages, and software designed for managing data, processes, and projects.

To view the list of Apps & Tools and their descriptions, readers are encouraged to refer to the [Apps & Tools Dashboard](#) on the [Jobs-Skills Portal](#).

Top 20 Apps & Tools in 2023 and Their Historical Rankings



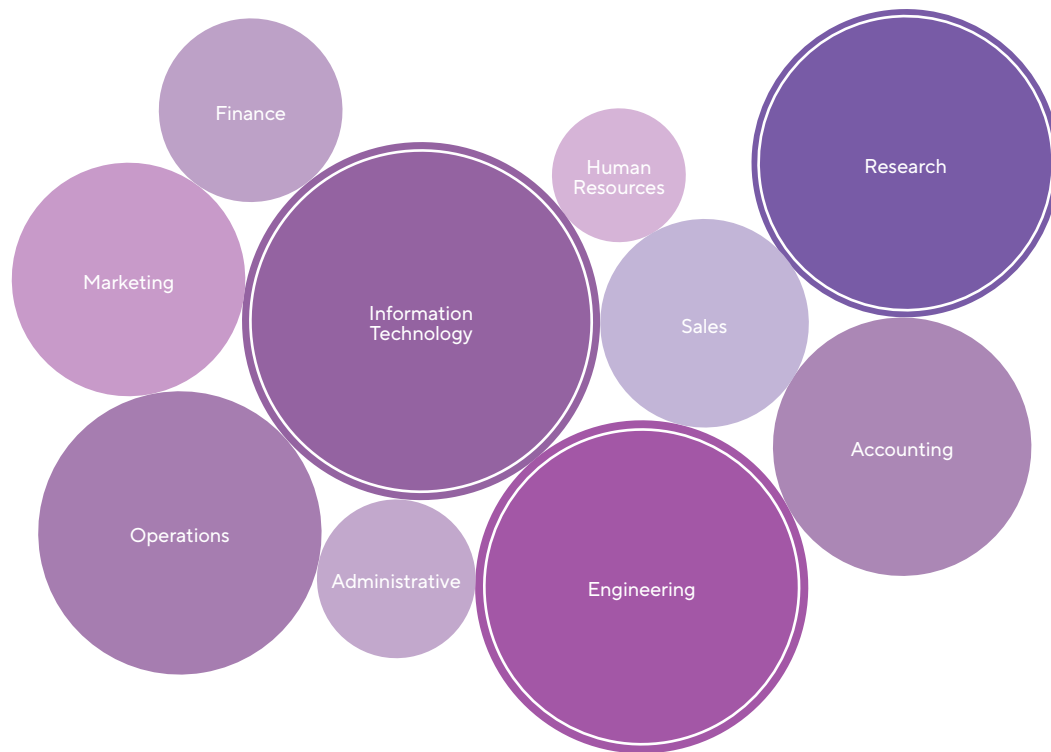
IT and Engineering were the top two work functions driving demand for both general Apps & Tools and those related to Artificial Intelligence (AI).

However, Research overtook Operations to be the third-largest work function to drive demand for AI-related Apps & Tools.

AI-related Apps & Tools are those that contribute to the creation, development, or deployment of AI, leverages AI capabilities, or associated with AI in general.

The top three work functions that use **AI-related Apps & Tools** are IT (29%), followed by Engineering (12%), and Research (7%), which make up 48% of all AI-related Apps & Tools required. Examples of AI-related Apps & Tools most required in these work functions are *Java*, *Python*, and *Google Analytics*, respectively.

Demand of AI-related Apps & Tools by Work Function in 2023



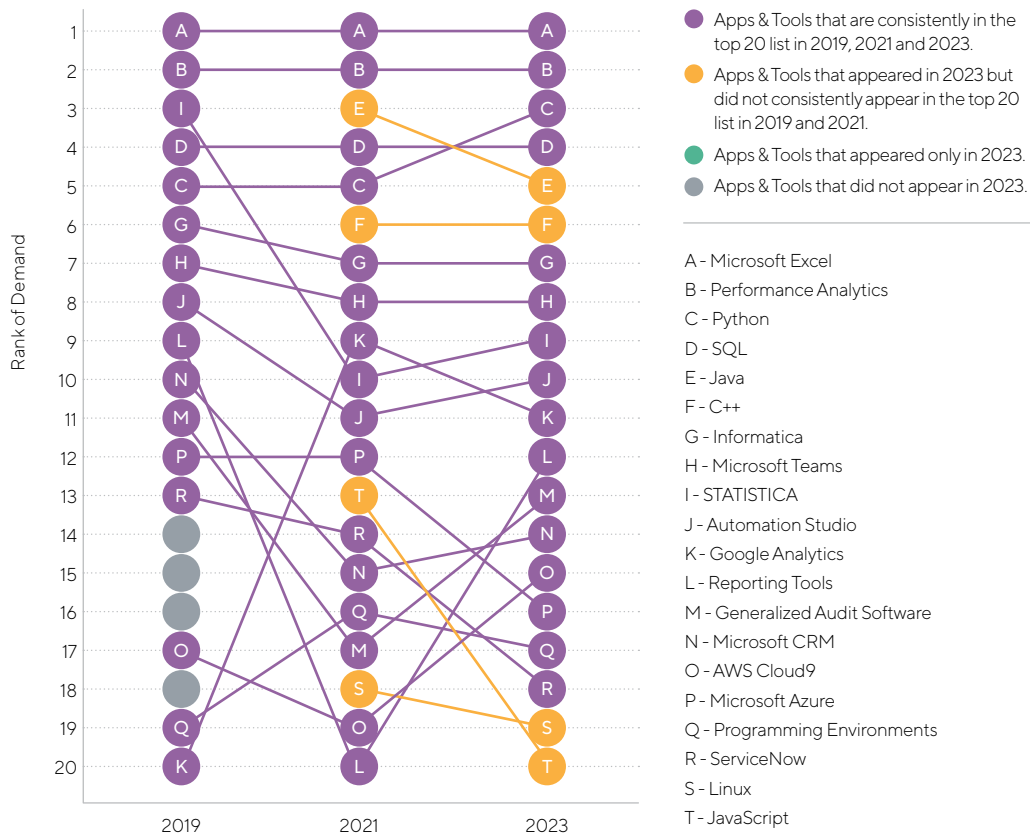
Most of the top 20 AI-related Apps & Tools had generally stable rankings between 2019 and 2023.

Some of the in-demand AI-related Apps & Tools were also among the top 20 Apps & Tools, such as *Microsoft Excel*, *Python*, and *SQL*.

Between 2019 and 2023, four of the fastest-changing AI-related Apps & Tools in demand were: *Microsoft CRM* (22nd to 14th), *Amazon Web Services Cloud9* (26th to 15th), *Microsoft Azure* (35th to 16th), and *ServiceNow* (23rd to 18th). The adoption of these Apps & Tools suggested more employers were embracing AI to enhance work processes and using cloud services in their digital transformation.

Please refer to the [Jobs-Skills Portal](#) for the list of AI-related Apps & Tools and their descriptions.

Top 20 AI-related Apps & Tools in 2023 and Their Historical Rankings



Using the Job Requirements Dashboard, readers may deep dive into the changing skill requirements of different job roles and analyse the use of Apps & Tools at the workplace. Three job roles were used to illustrate the changing job requirements using the [Job Requirements Dashboard](#).

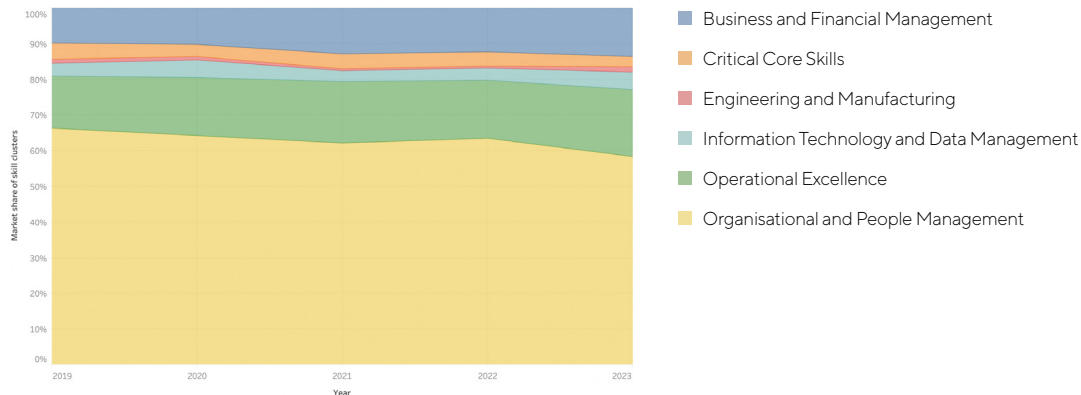
The skill requirements for social workers have shifted to place greater emphasis on skills that involve analysis, outcome evaluation, and implementation of programmes.

Changes to the skill requirements reflected a stronger focus on enhancing service delivery through strategic and data-driven methods.

As the demand for the biggest skill cluster diminishes, it becomes more important for social workers to develop skills in smaller skill clusters in response to changing business needs.

Demand for social workers² has increased, driven by Singapore's ageing population and increased awareness of social programmes.

Skill Clusters of Social Workers



Information for this job role:

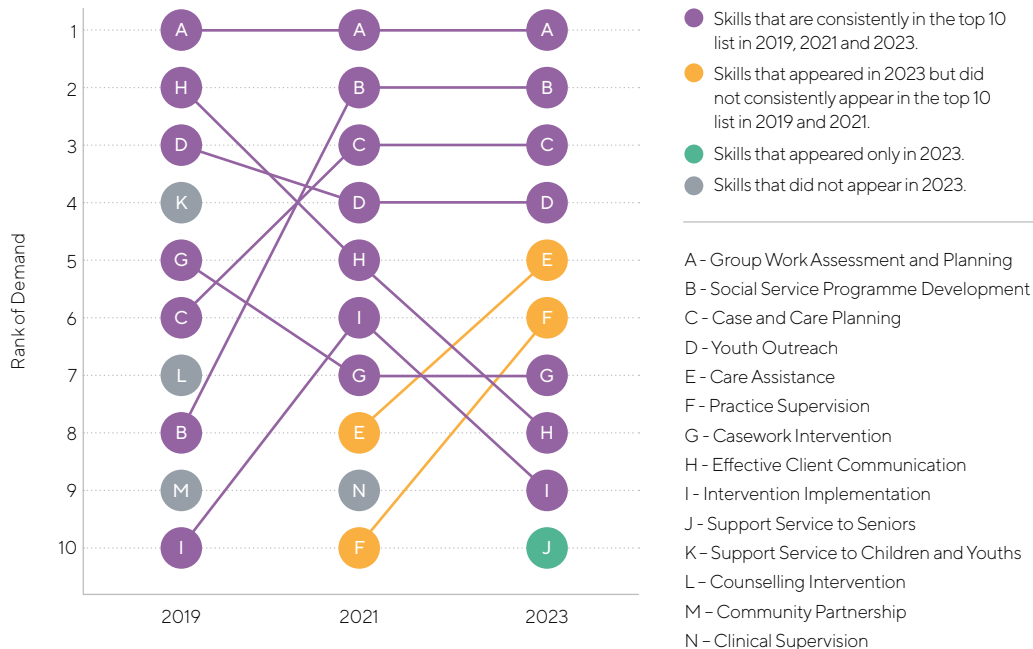
- +21% Change in Job Demand (Growth from 2020 to 2023)
- 0.1% 2023 Job Demand
- S\$4,100 Salary (Gross Median in 2023)
- +S\$400 Difference in Salary (Gross Median in 2023 and 2020)

² Based on Singapore Standard Occupation Classification (SSOC) code 26351: 'Social worker (general)'.

'Organisational and People Management', the largest skill cluster, decreased to 58% in 2023.

Between 2019 and 2023, seven skills consistently appeared in the top 10 ranks within this cluster. *Group Work Assessment and Planning* remained the top skill across 2019, 2021, and 2023. *Social Service Programme Development*, *Case and Care Planning*, *Care Assistance*, and *Support Service to Seniors* were skills that gained importance in 2023.

Top 10 Skills within 'Organisational and People Management' for Social Workers

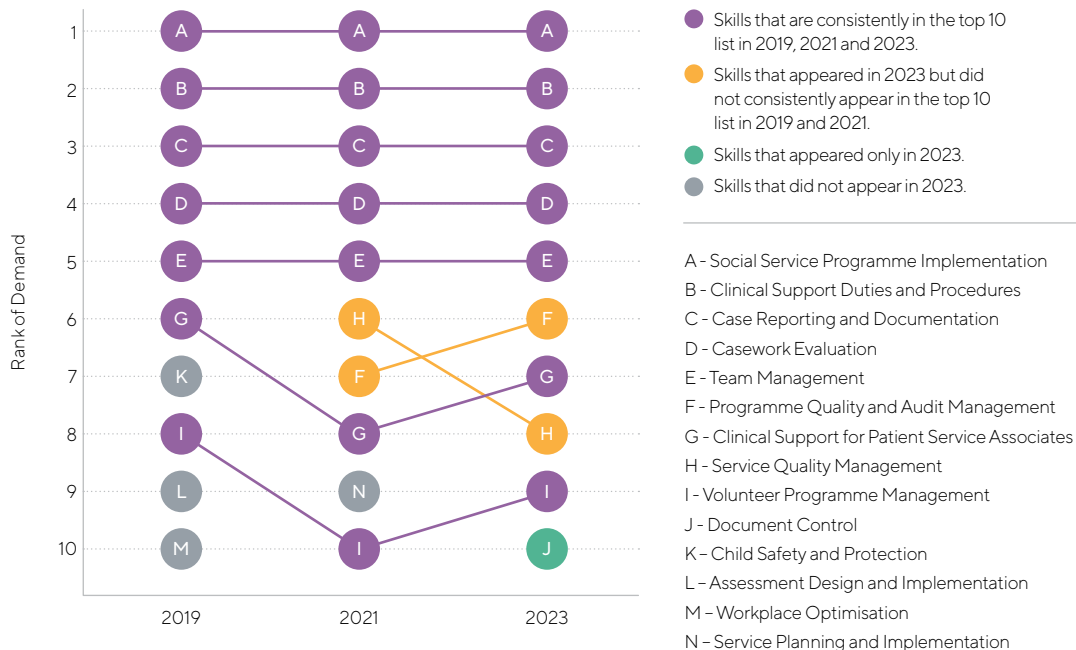




'Operational Excellence', the second-largest skill cluster, increased to 19% in 2023.

Between 2019 and 2023, seven skills consistently appeared in the top 10 ranks within this cluster. The top four skills maintained their spots across 2019, 2021, and 2023. *Programme Quality and Audit Management* and *Document Control* were skills that gained importance in 2023.

Top 10 Skills within 'Operational Excellence' for Social Workers

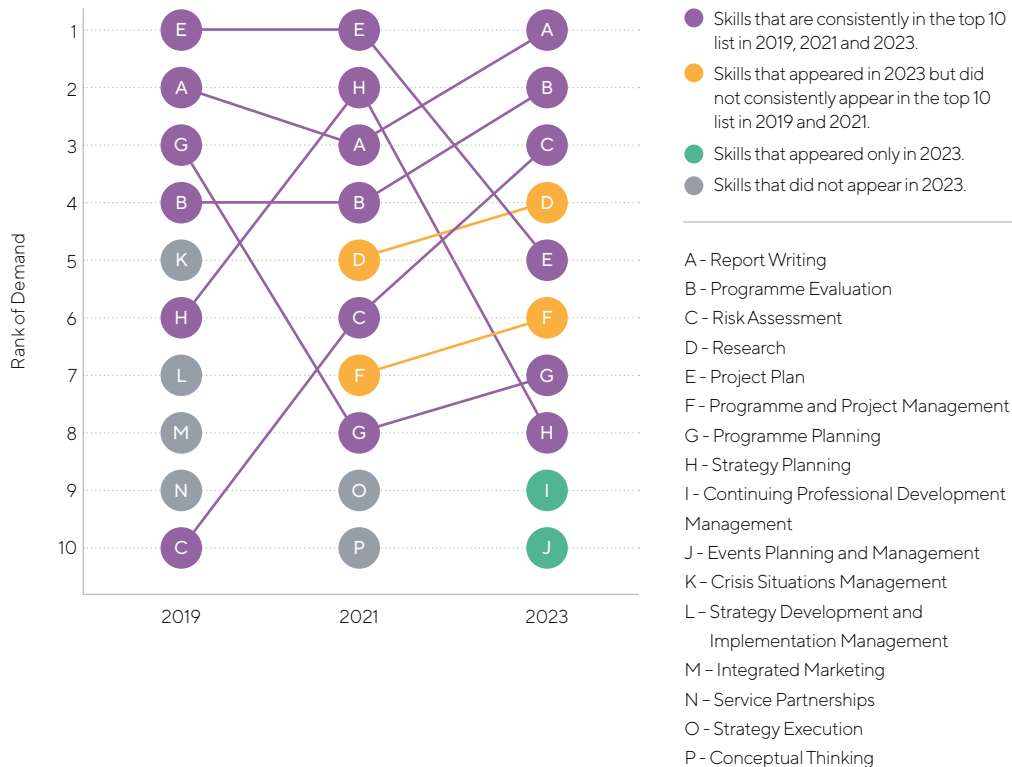




'Business and Financial Management', the third-largest skill cluster, increased to 14% in 2023.

Between 2019 and 2023, six skills consistently appeared in the top 10 ranks within this cluster. *Project Plan* – the top skill for 2019 and 2021 – dropped in importance in 2023 and was replaced by the skills *Report Writing* and *Programme Evaluation*. *Risk Assessment*, *Research*, and *Programme and Project Management* were skills that gained importance in 2023.

Top 10 Skills within 'Business and Financial Management' for Social Workers

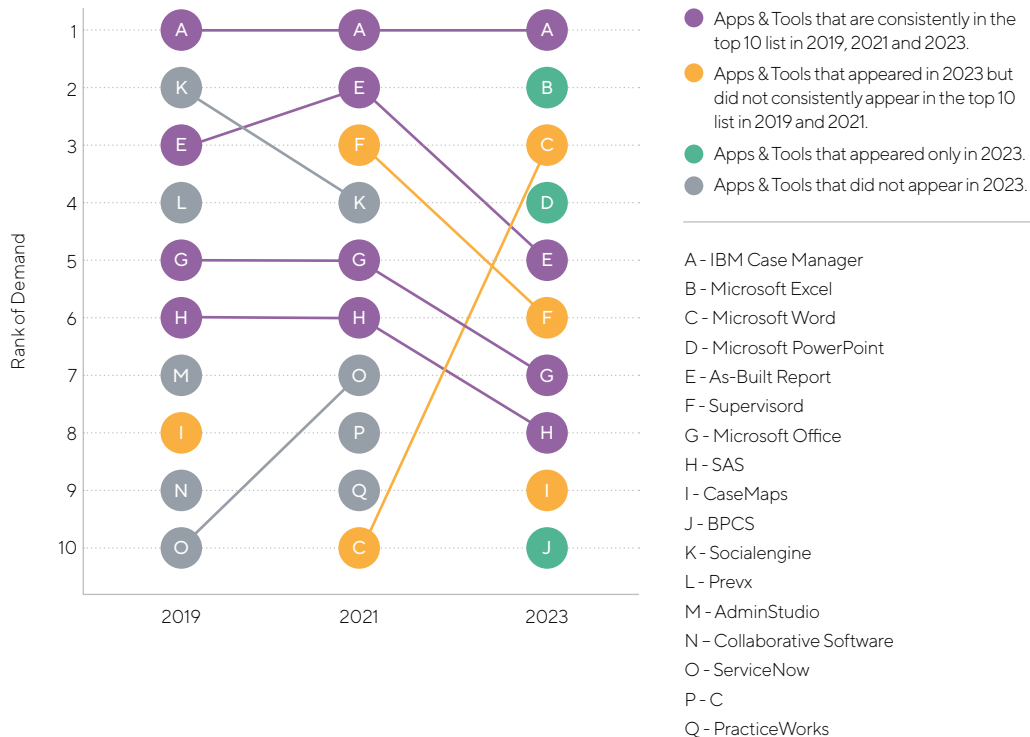




Social workers generally used a different set of Apps & Tools between 2019 and 2023, as only four Apps & Tools remained in the top 10 over the years.

Between 2019 and 2023, four Apps & Tools consistently appeared in the top 10 ranks. *IBM Case Manager* consistently remained the top Apps & Tools across 2019, 2021, and 2023. *Microsoft PowerPoint*, *Microsoft Excel*, and *BPCS* gained importance in 2023. *SAS* was consistently among the overall top 10 Apps & Tools demanded in job postings for social workers. This statistical software suite has AI capabilities to help with analysis, decision-making, and optimisation solutions.

Top 10 Apps & Tools for Social Worker





Skill requirements for marketing and sales executives remained relatively stable over the years.

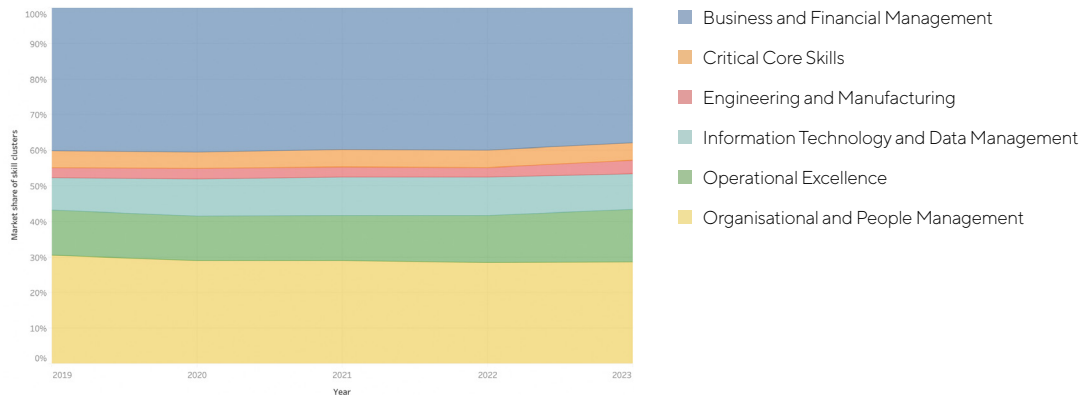
The increasing demand for the digitalisation of operations and the focus on sustainability are shaping the skill requirements.

Marketing and sales executives can continue to deepen mastery in their main skill clusters, while remaining agile and ready to adapt to emerging trends in other skill clusters.

These professionals require skills in two main skill clusters, namely 'Business and Financial Management' (38% in 2023), and 'Organisational and People Management' (29% in 2023).

Demand for marketing and sales executives³ increased between 2019 and 2023.

Skill Clusters of Marketing and Sales Executives



Information for this job role:

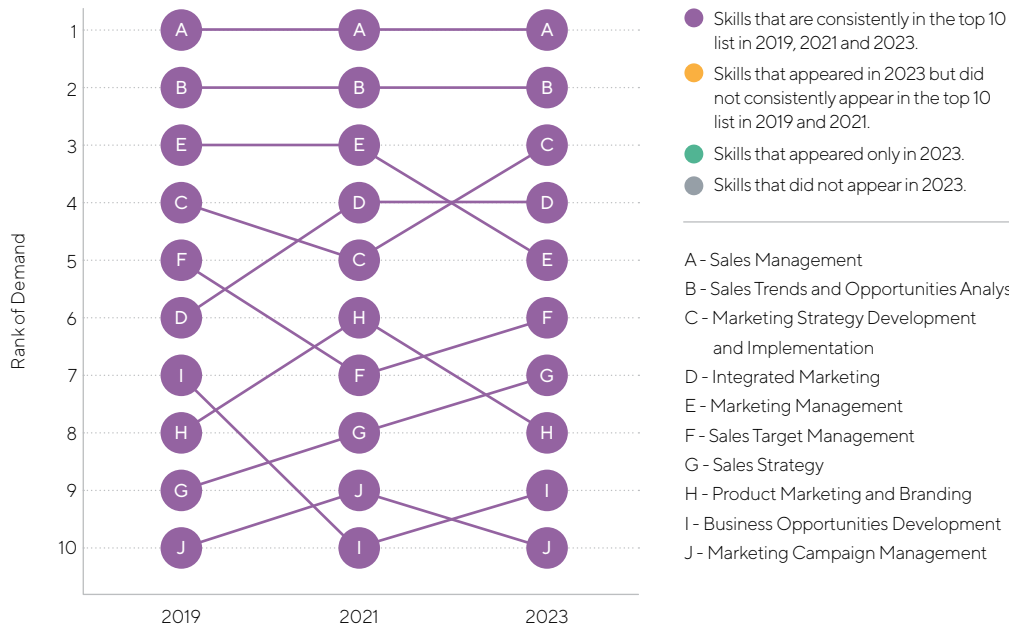
- +0.9% Change in Job Demand (Growth from 2020 to 2023)
- 0.2% 2023 Job Demand
- S\$5,000 Salary (Gross Median in 2023)
- +S\$700 Difference in Salary (Gross Median in 2023 and 2020)

³ Based on Singapore Standard Occupation Classification (SSOC) code 33223: 'Marketing and sales executive (food & beverage services)'.

'Business and Financial Management', the largest skill cluster, decreased to 38% in 2023.

Between 2019 and 2023, all 10 skills maintained their top 10 ranks within this cluster. *Sales Management* and *Sales Trends and Opportunities Analysis* were consistently the top two skills across 2019, 2021, and 2023. *Integrated Marketing* and *Sales Strategy* gained importance in 2023.

Business and Financial Management Skill Cluster

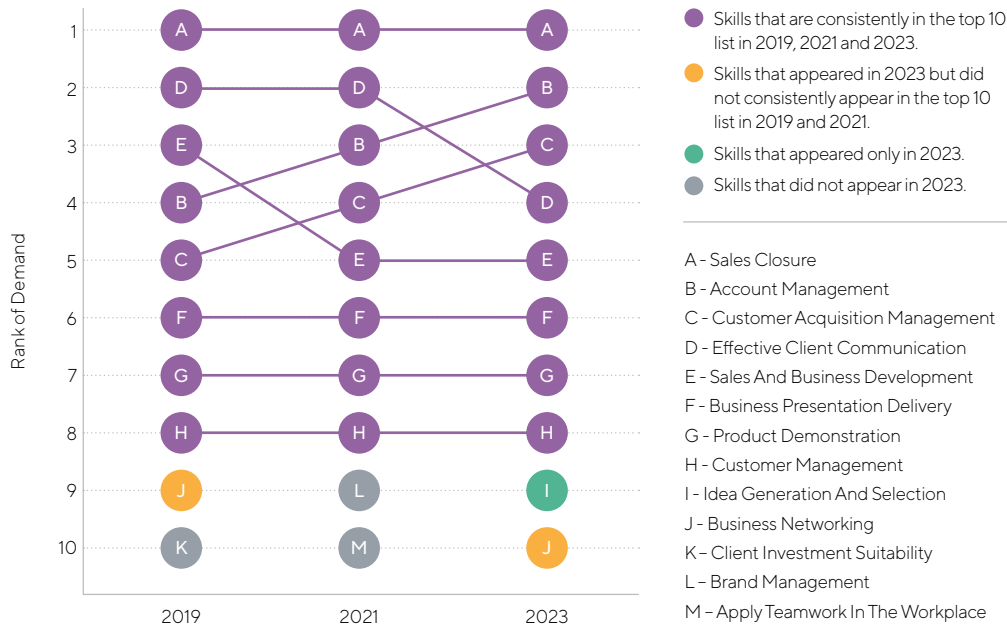




'Organisational and People Management', the second-largest skill cluster, decreased to 29% in 2023.

Between 2019 and 2023, eight skills consistently appeared in the top 10 ranks within this cluster. *Sales Closure* remained the top skill across 2019, 2021, and 2023. *Account Management*, *Customer Acquisition Management*, and *Idea Generation and Selection* gained importance in 2023.

Organisational and People Management Skill Cluster

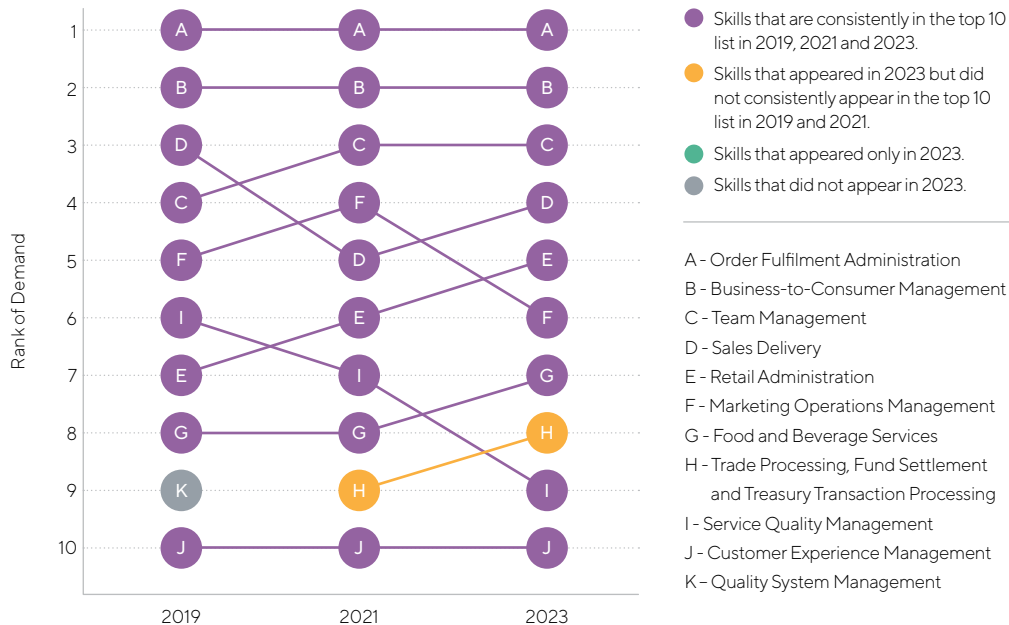




'Operational Excellence', the third-largest skill cluster, increased to 15% in 2023.

Between 2019 and 2023, nine skills consistently appeared in the top 10 ranks within this cluster. *Order Fulfilment Administration* and *Business-to-Consumer Management* maintained their top two spots across 2019, 2021, and 2023. *Team Management*, *Retail Administration*, *Food and Beverage Services*, and *Trade Processing, Fund Settlement and Treasury Transaction Processing* were skills that gained importance in 2023.

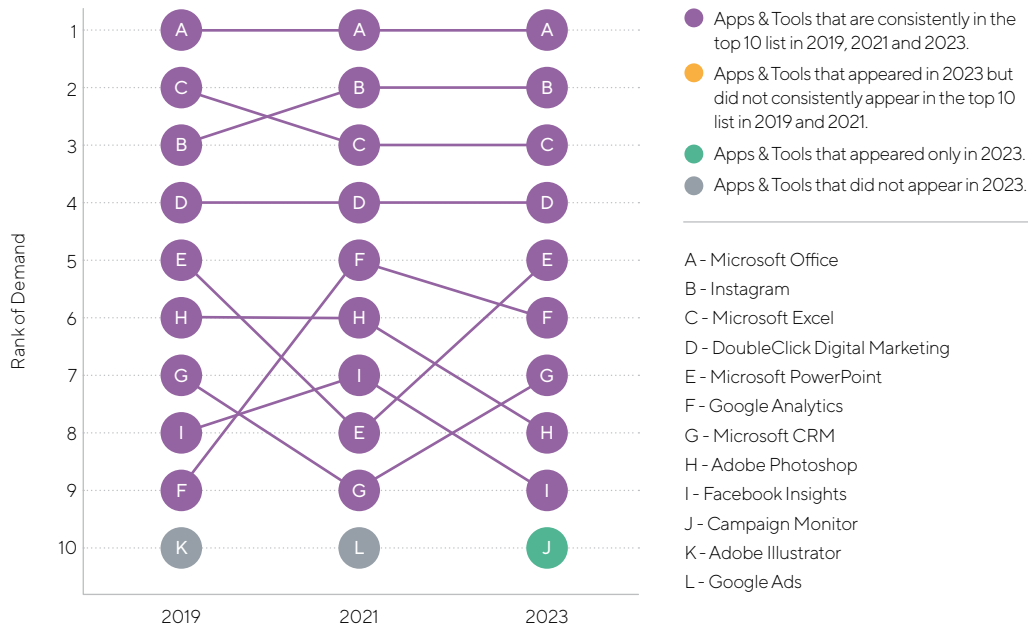
Operational Excellence Skill Cluster



Apps and Tools used by marketing and sales executives between 2019 and 2023 remained largely unchanged, including the maturing use of social media channels and platforms for marketing and sales generation.

Between 2019 and 2023, nine Apps & Tools consistently appeared in the top 10 ranks. *Microsoft Office* consistently topped the Apps & Tools list across 2019, 2021, and 2023. *Adobe Photoshop* was among the top Apps & Tools identified in job postings for marketing and sales executives. This image editing software suite integrates Generative Artificial Intelligence (GenAI) to help with editing objects and the background, and re-rendering objects.

Top 10 Apps & Tools For Marketing and Sales Executives



The skill requirements for resident technical officers shifted to place greater emphasis on skills that involved resource coordination and timely execution, as well as overseeing work quality and collaboration with external parties.

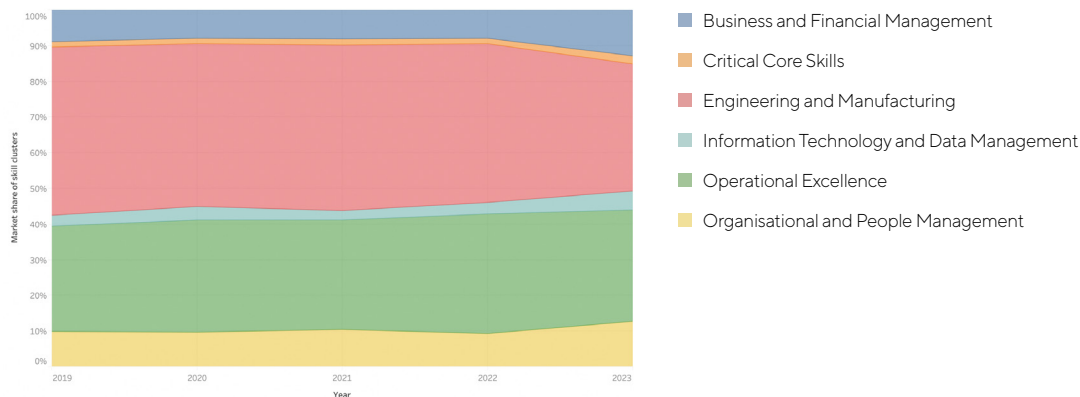
Changes to the skill requirements reflected stronger focus on managerial responsibilities, with less emphasis on technical expertise in the 'Engineering and Manufacturing' cluster.

Resident technical officers have skills in two main skill clusters – 'Engineering and Manufacturing' (36% in 2023) and 'Operational Excellence' (31% in 2023).

While the barrier to acquiring skills in their main skill clusters may be lower due to existing expertise, market trends suggest that resident technical officers will need to develop skills in other clusters as well.

Demand for resident technical officers⁴ increased between 2019 and 2023, driven by steady improvement in construction demand over the medium term. Wages rose substantially to tackle the skilled manpower shortage⁵ and to attract qualified professionals amidst increased construction activity.

Skill Clusters of Resident Technical Officers



Information for this job role:

- +11% Change in Job Demand (Growth from 2020 to 2023)
- 2% 2023 Job Demand
- S\$4,400 Salary (Gross Median in 2023)
- +S\$1,800 Difference in Salary (Gross Median in 2023 and 2020)

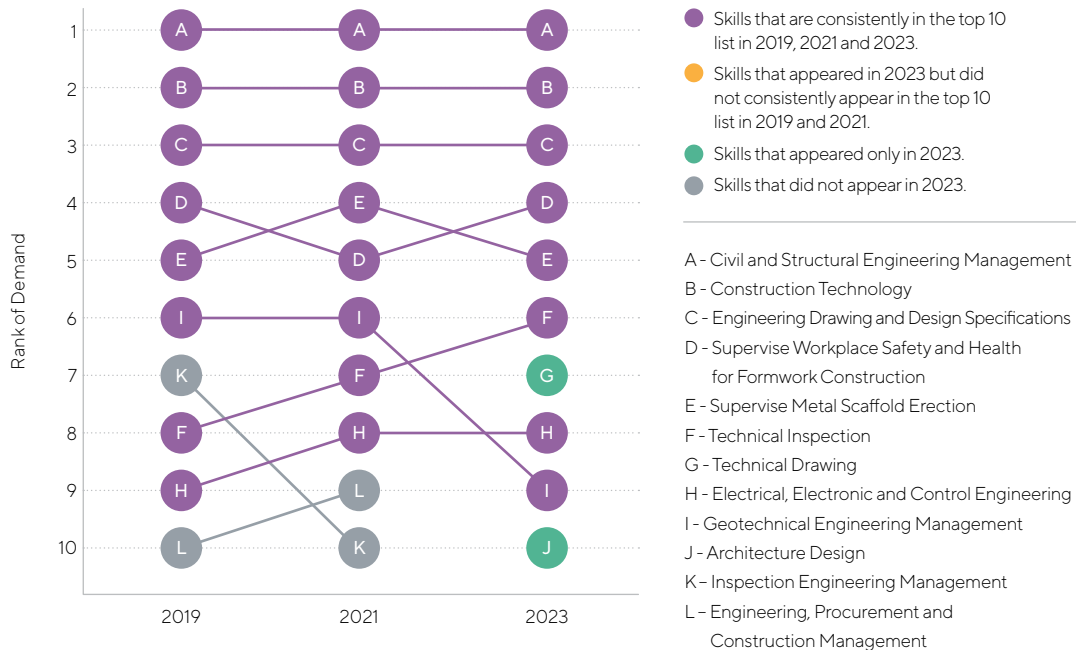
⁴ Based on Singapore Standard Occupation Classification (SSOC) code 31124: 'Resident technical officer'.

⁵ The Business Times, 2024.

'Engineering and Manufacturing' cluster decreased to 36% in 2023, but remains as the dominant skill cluster.

Between 2019 and 2023, eight skills consistently appeared in the top 10 ranks within this cluster. *Civil and Structural Engineering Management*, *Construction Technology* and *Engineering Drawing and Design Specifications* remained the top three skills across 2019, 2021, and 2023. *Technical Inspection*, *Electrical, Electronic and Control Engineering*, *Technical Drawing* and *Architecture Design* were skills that gained importance in 2023.

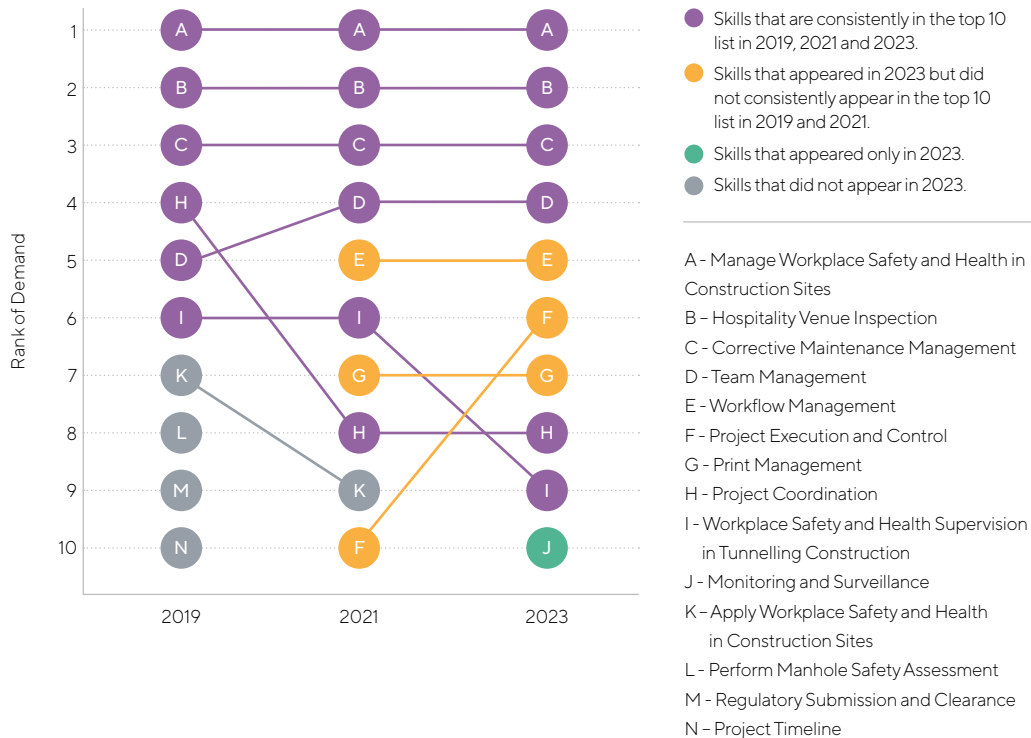
Engineering and Manufacturing Skill Cluster



'Operational Excellence', the second-largest skill cluster, increased to 31% in 2023.

Between 2019 and 2023, six skills consistently appeared in the top 10 ranks within this cluster. *Manage Workplace Safety and Health in Construction Sites*, *Hospitality Venue Inspection* and *Corrective Maintenance Management* maintained their top spots across 2019, 2021, and 2023. *Team Management*, *Project Execution and Control*, and *Monitoring and Surveillance* were skills that gained importance in 2023.

Operational Excellence Skill Cluster

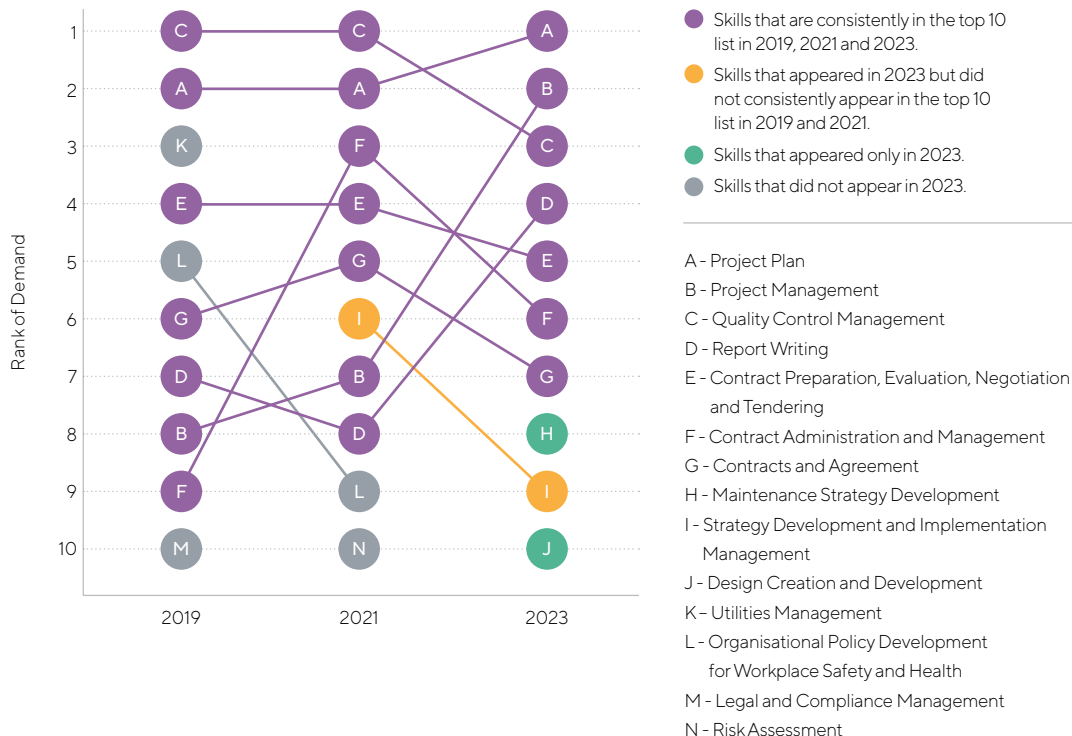




'Business and Financial Management', the third-largest skill cluster, increased to 13% in 2023.

Between 2019 and 2023, seven skills consistently appeared in the top 10 ranks within this cluster. *Project Plan* overtook *Quality Control Management* to become the top skill in 2023. *Project Management*, *Report Writing*, *Maintenance Strategy Development*, and *Design Creation and Development* were skills that gained importance in 2023.

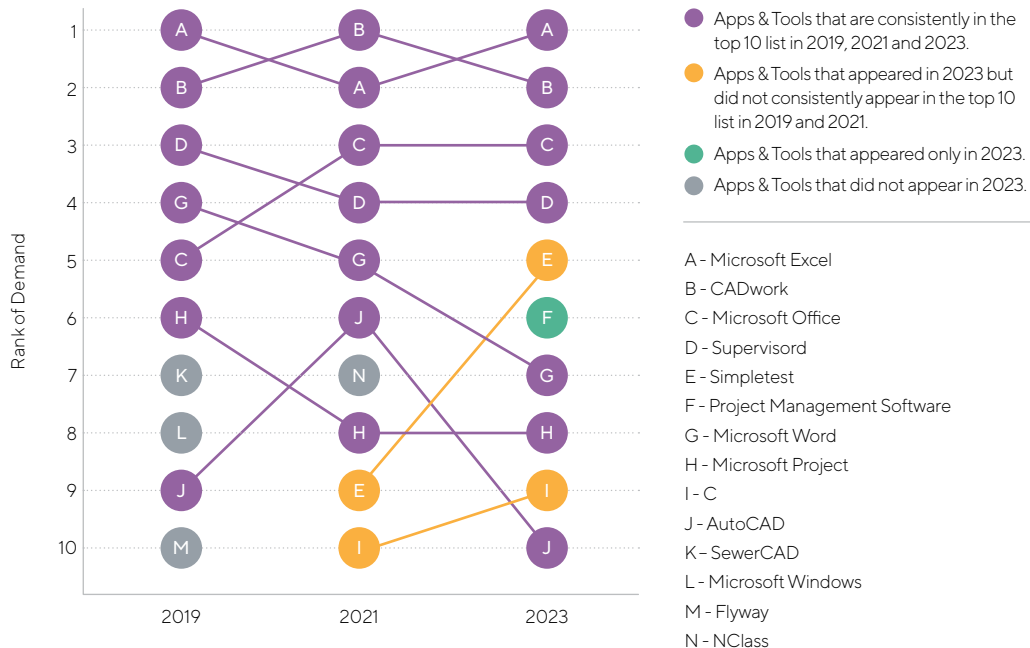
Business and Financial Management Skill Cluster



Resident technical officers generally used a different set of Apps & Tools between 2019 and 2023, as only seven Apps & Tools remained in the top 10 over the years.

Between 2019 and 2023, seven Apps & Tools consistently appeared in the top 10 ranks. *Microsoft Excel* and *CADwork* remained the top two Apps & Tools, although their rankings switched in 2023. *Microsoft Office*, *Simpletest*, *Project Management Software*, and *C* were Apps & Tools that gained importance in 2023. *ServiceNow* was observed as one of the top AI-related Apps & Tools required for resident technical officers. This enterprise software platform can be used for project management and reporting. It leverages AI to support functions such as performance analytics and process optimisation⁶.

Top 10 Apps & Tools for Residential Technical Officers



⁶ ServiceNow, accessed 2024.

CHAPTER 2

Priority Skills Forecast and Movement

In this chapter, the movement trends of the updated set of **Priority skills** from the **Care, Digital, and Green economies** were analysed using historical data from 2022 to 2024. A four-quadrant Skills Matrix was used to enhance the insights on the change in skills demand and transferability. The analysis also forecasted the demand of the Priority Skills in the next 12 months. Finally, the chapter concludes by highlighting eleven skills to watch for in the three economies.

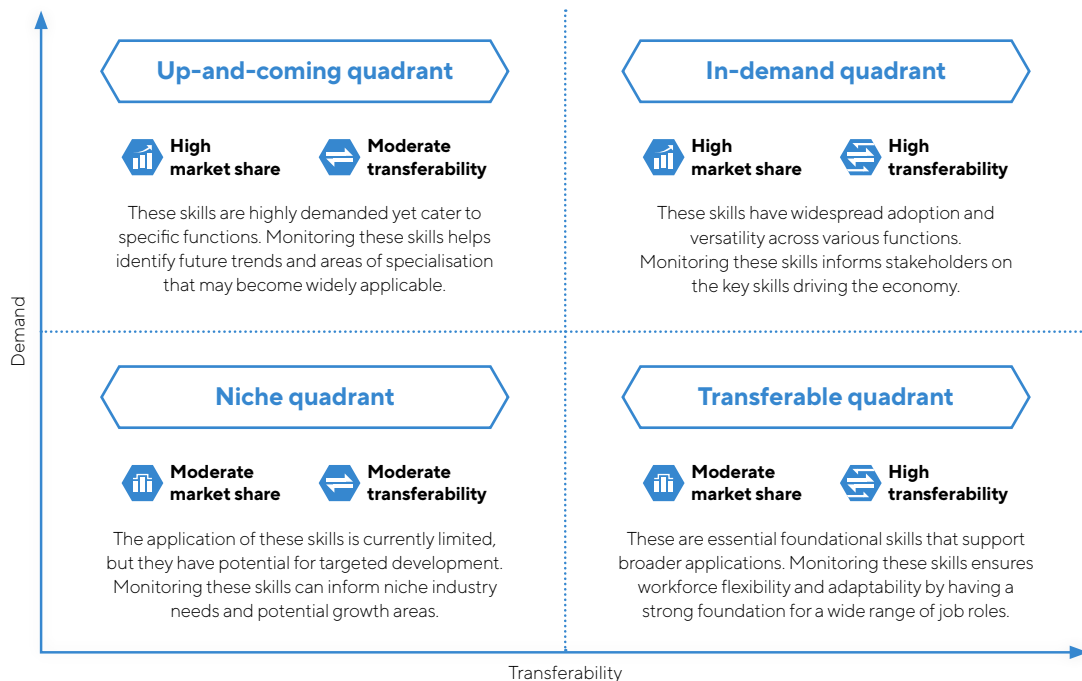
The [Priority Skills Dashboard](#) is used to generate the insights for this chapter and is available on the [Jobs-Skills Portal](#) for readers to explore further insights.

Click [here](#) to watch videos showcasing our partners' perspectives.

Priority skills were first plotted before analysing skills movements for all three economies.

Click [here](#) to access the full list of Priority Skills in respective quadrant in Care, Digital and Green economies.

Skills Matrix



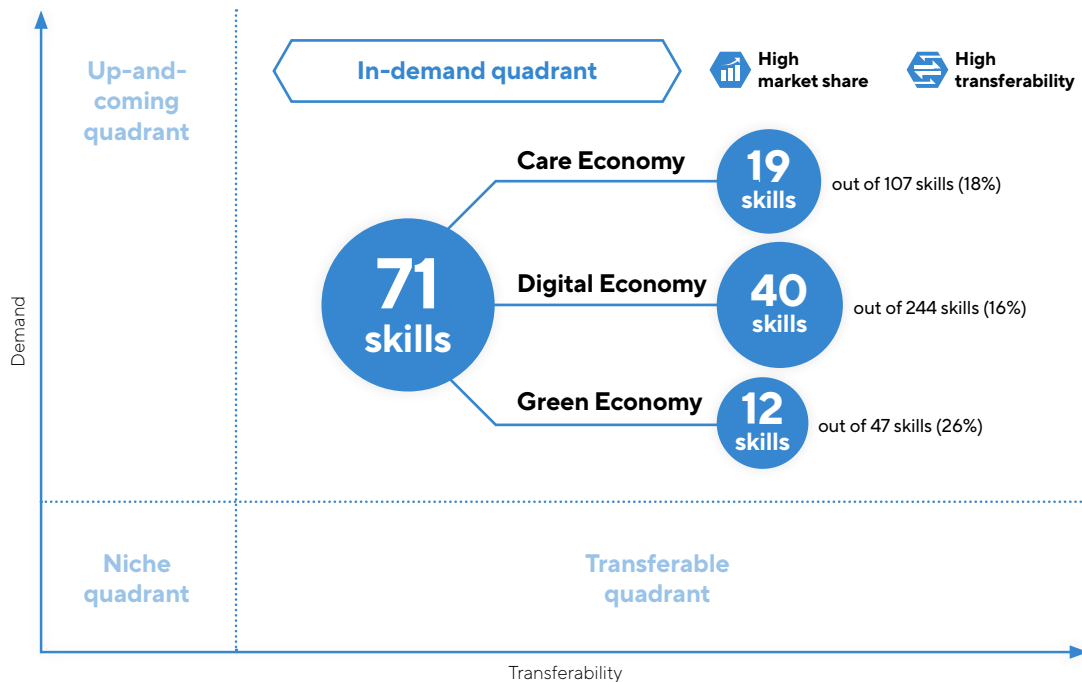
The Skills Matrix is a framework that locates the momentum of change of respective priority skills based on their demand and transferability trends. A data point, for each year of a given skill, was categorised into either In-demand quadrant, Up-and-coming quadrant, Niche quadrant, or Transferable quadrant. This categorisation provides readers a clearer perspective of skills demand and transferability tempo across the years.



Observation One: Skills that were consistently in the In-demand quadrant

71 skills were consistently in high demand and were highly transferable across the three years. This trend is expected to continue into 2025.

From 2022 to 2024, a total of 71 skills, or 18% of 398 skills across the three economies, maintained their position in the In-demand quadrant. This trend is expected to continue into 2025. Skills that exhibit such characteristics – high demand and transferability – remain highly sought-after due to business demand.





Observation One: Skills that were consistently in the In-demand quadrant

Within the Care Economy, 19 skills were in the In-demand quadrant from 2022 to 2024. This trend is expected to continue into 2025.

19 skills, or about 18% of 107 skills in this economy, were identified to be in the In-demand quadrant in the past three years. Among them, two were selected for illustration.

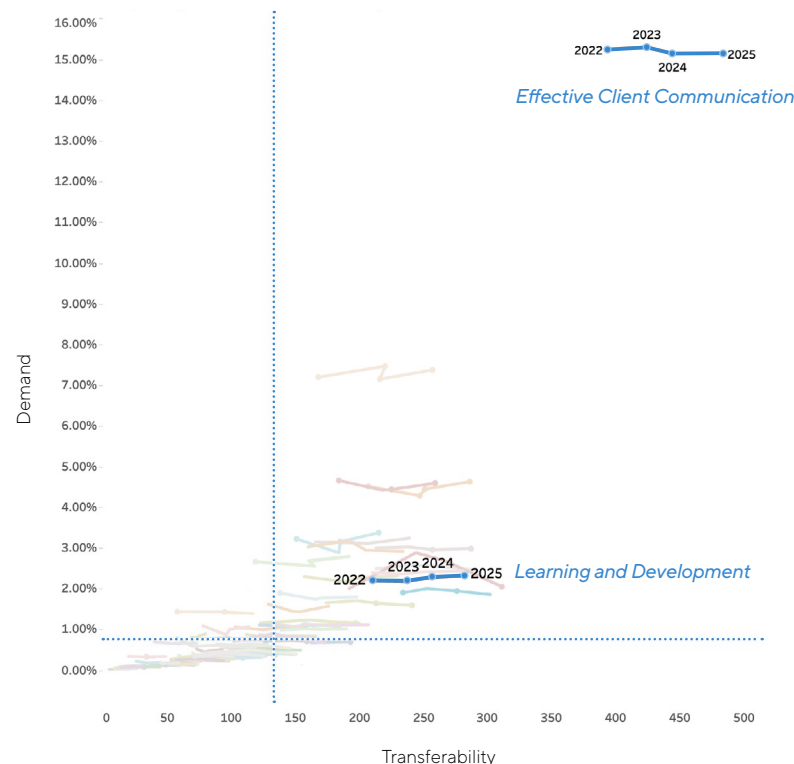
Effective Client Communication

This skill is important in job roles like administration managers, accountants, and marketing managers. It is also essential in effective healthcare services delivery to ensure up-to-date physical and digital communication channels¹.

Learning and Development

Job roles requiring this skill include preschool teachers, web and mobile application developers, and administration managers. This skill has increased in importance due to greater adoption of EdTech platforms, enabling borderless learning, and GenAI chatbots, accelerating the pace of learning².

Movement of Priority Skills in the Care Economy related to Observation One



¹ Kanika Vats, National Library of Medicine, 2024.

² Claude Jeffrey Renaud et. al, Yong Loo Lin School of Medicine, 2020.



Observation One: Skills that were consistently in the In-demand quadrant

Within the Digital Economy, 40 skills were in the In-demand quadrant from 2022 to 2024. This trend is expected to continue into 2025.

40 skills, or about 16% of 244 skills in this economy, were identified to be in the In-demand quadrant in the past three years. Among them, two were selected for illustration.

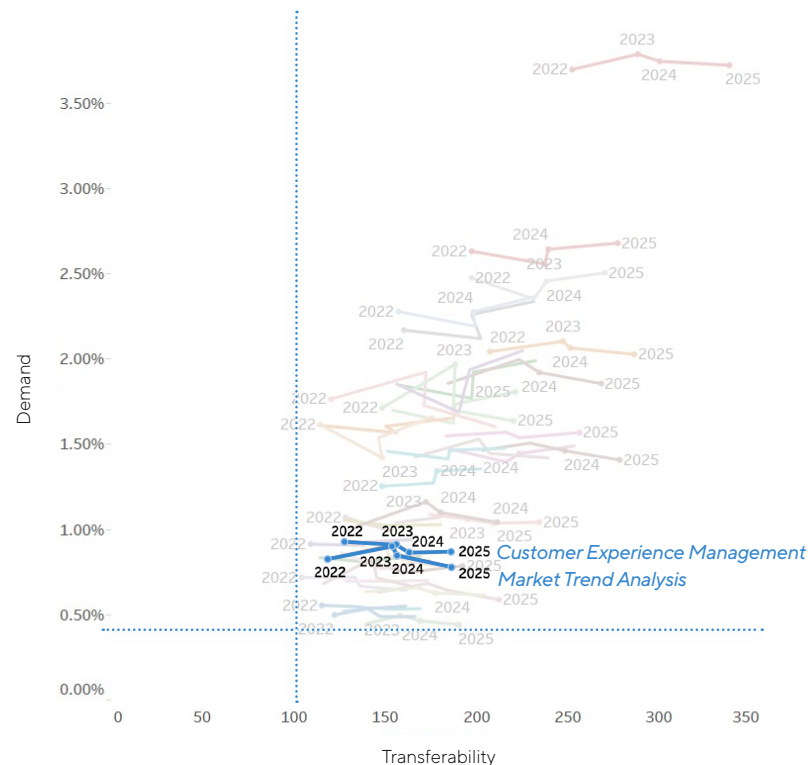
Customer Experience Management

Job roles requiring this skill include customer service officers, marketing managers and after-sales advisers. This skill is increasingly important as businesses pursue personalised customer service³ and quality customer experiences to differentiate themselves from their competitors⁴.

Market Trend Analysis

Job roles requiring this skill include marketing managers, marketing and sales executives, and business development executives. This is a skill required to uncover deeper insights about customers, competitors, and market trends.

Movement of Priority Skills in the Digital Economy related to Observation One



³ KPMG, 2019

⁴ McKinsey, 2023.



Observation One: Skills that were consistently in the In-demand quadrant

Within the Green Economy, 12 skills were in the In-demand quadrant from 2022 to 2024. This trend is expected to continue into 2025.

12 skills, or about 26% of 47 skills, in this economy were identified to be in the In-demand quadrant in the past three years. Among them, two were selected for illustration.

Sustainability Management

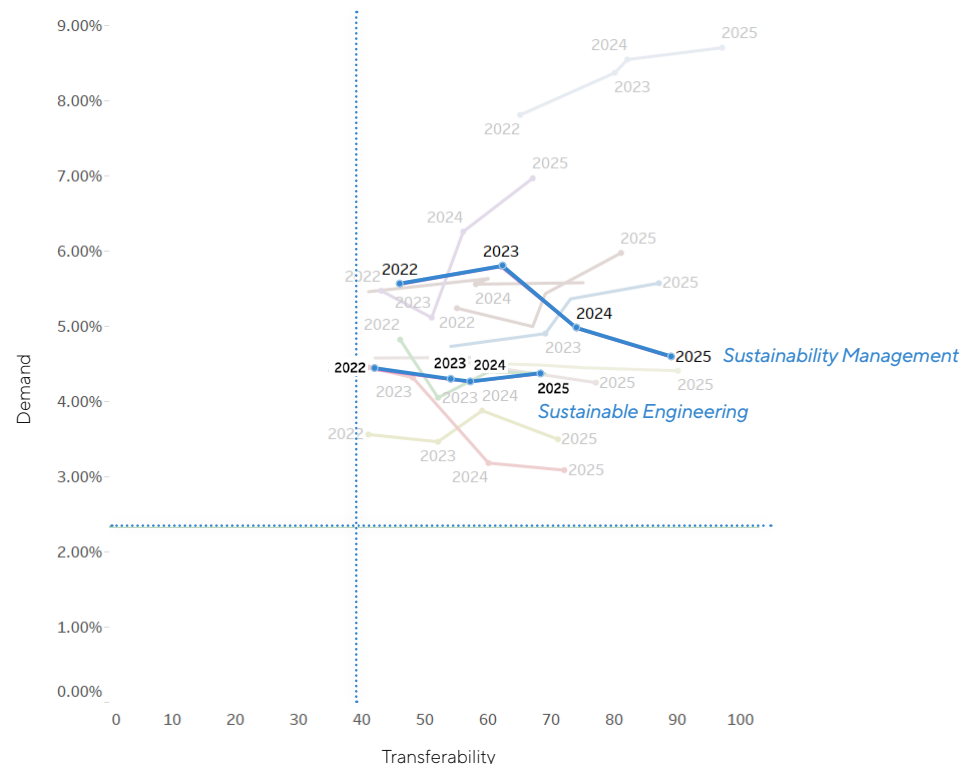
Job roles requiring this skill include premises and facilities maintenance managers, business consultants, and business and financial project managers. This skill is crucial for organisational strategies, resource utilisation assessments and identifying efficiency improvements.

Sustainable Engineering

Business and financial project managers, mechanical engineers, and environmental engineers are some job roles that demand this skill. This skill is vital for designing and operating systems that optimise energy use and environmental performance.

The Singapore Green Plan 2030⁵ continues to drive sustained demand for skills that collectively support sustainability initiatives in the green economy.

Movement of Priority Skills in the Green Economy related to Observation One



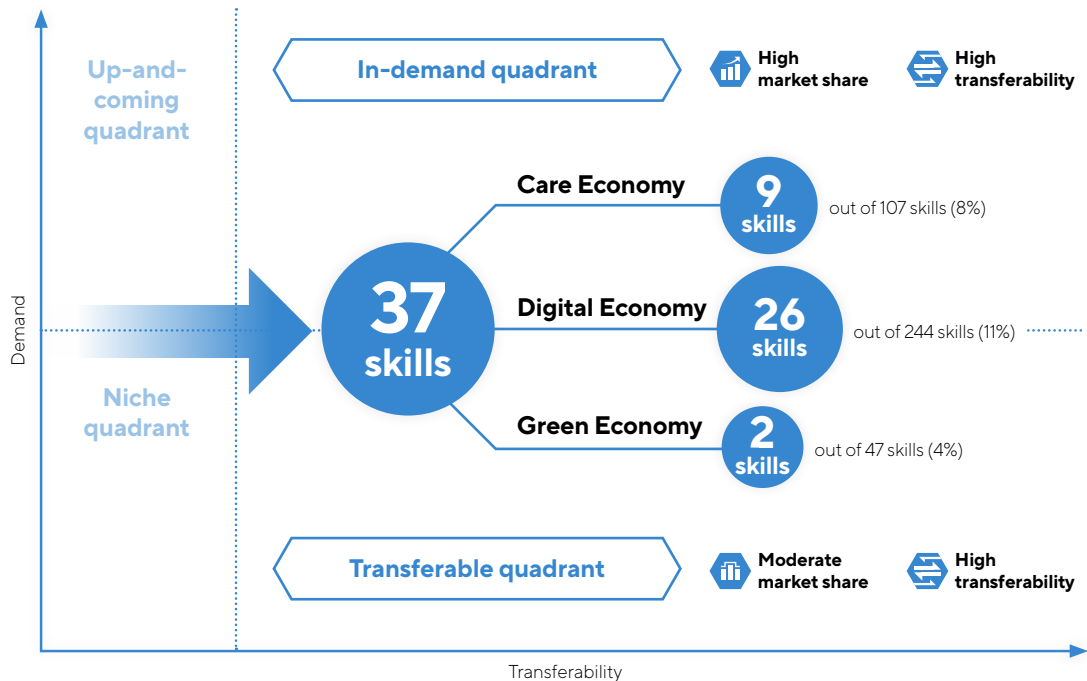
⁵ Singapore Green Plan 2030, 2024.



Observation Two: Skills that moved from moderate to high transferability

37 skills moved from moderate to high transferability in 2024. This trend is expected to continue into 2025.

A total of 37 skills, or about 9% of 398 skills across the three economies, crossed the 70th percentile mark between 2022 and 2023, and were in the In-demand and Transferable quadrants in 2024. This trend is expected to continue into 2025. While majority of skills were generally increasing in transferability, these identified skills were more widely adopted compared to other priority skills across various job roles.





Observation Two: Skills that moved from moderate to high transferability

Within the Care Economy, 9 skills moved from moderate to high transferability.

9 skills, or about 8% of 107 skills, crossed the 70th percentile mark between 2022 and 2023 in terms of transferability, attributable to an increase in skills application across job roles in the economy. Among them, two were selected for illustration.

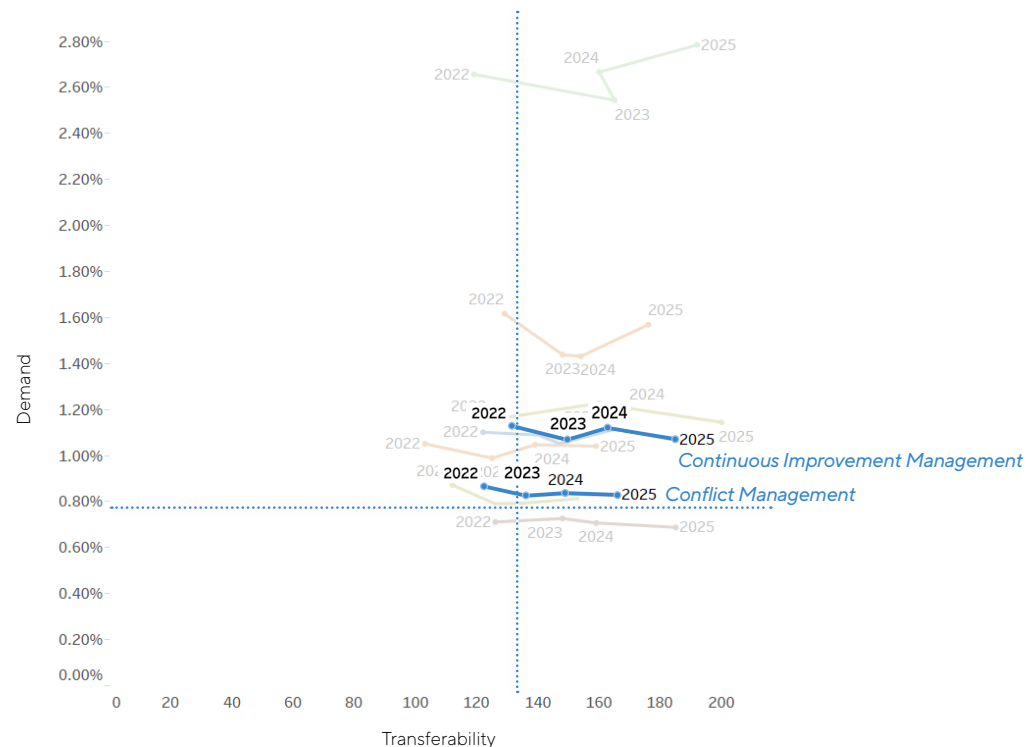
Conflict Management

Job roles requiring this skill include project managers, HR managers, and business consultants. This skill is increasingly important as workplace conflicts are on the rise, driven by labour market globalisation, economic fluctuations, and organisational restructuring⁶.

Continuous Improvement Management

This skill is demanded in job roles such as logistics or production planners, accountants, and procurement managers. This skill is gaining importance as more companies are adopting AI to improve business efficiency⁷.

Movement of Priority Skills in the Care Economy related to Observation Two



⁶ Giovanna Castellini et. al, National Library of Medicine, 2023.

⁷ Pricewaterhouse Coopers (PwC), 2024.



Observation Two: Skills that moved from moderate to high transferability

Within the Digital Economy, 26 skills moved from moderate to high transferability.

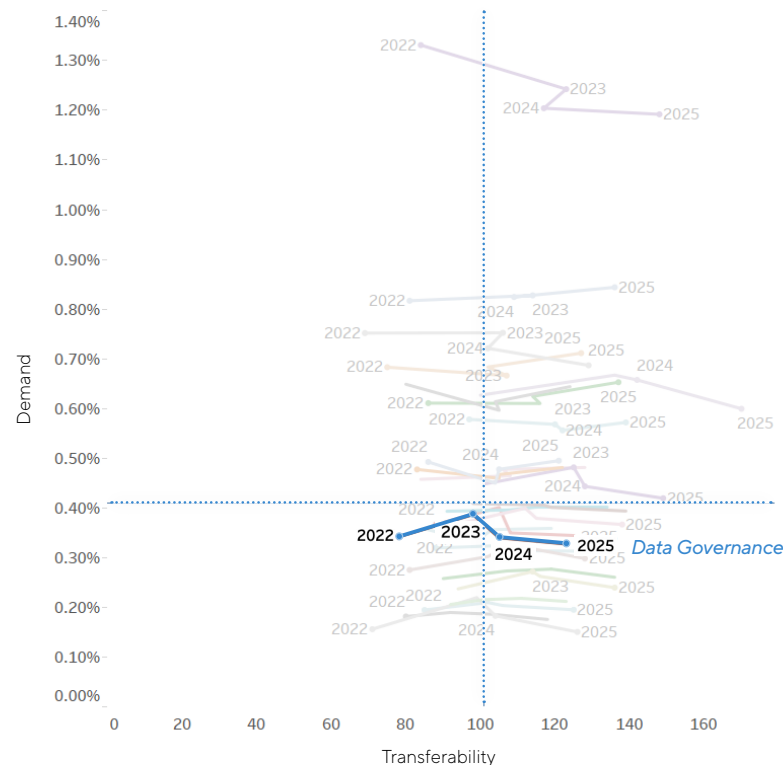
26 skills, or about 11% of 244 skills, crossed the 70th percentile mark between 2022 and 2023 in terms of transferability, attributable to an increase in skills application across job roles in the economy. Among them, *Data Governance* was selected for illustration.

This skill progressed from the Niche quadrant to the Transferable quadrant and is projected to remain so in 2025.

Job roles requiring this skill include data analysts, database architects, and business consultants. Enterprises have become reliant on data to inform decision-making, processes, and operations. As data protection regulations and enforcement become more stringent over the years⁸, this skill is needed for organisations to keep up with these evolving standards and frameworks.

With the prevalent use of data in business that permeates across the organisation, even non-tech job roles such as sales managers, financial derivatives dealers, and hotel operations managers will increasingly require this skill.

Movement of Priority Skills in the Digital Economy related to Observation Two





Observation Two: Skills that moved from moderate to high transferability

Within the Green Economy, 2 skills moved from moderate to high transferability.

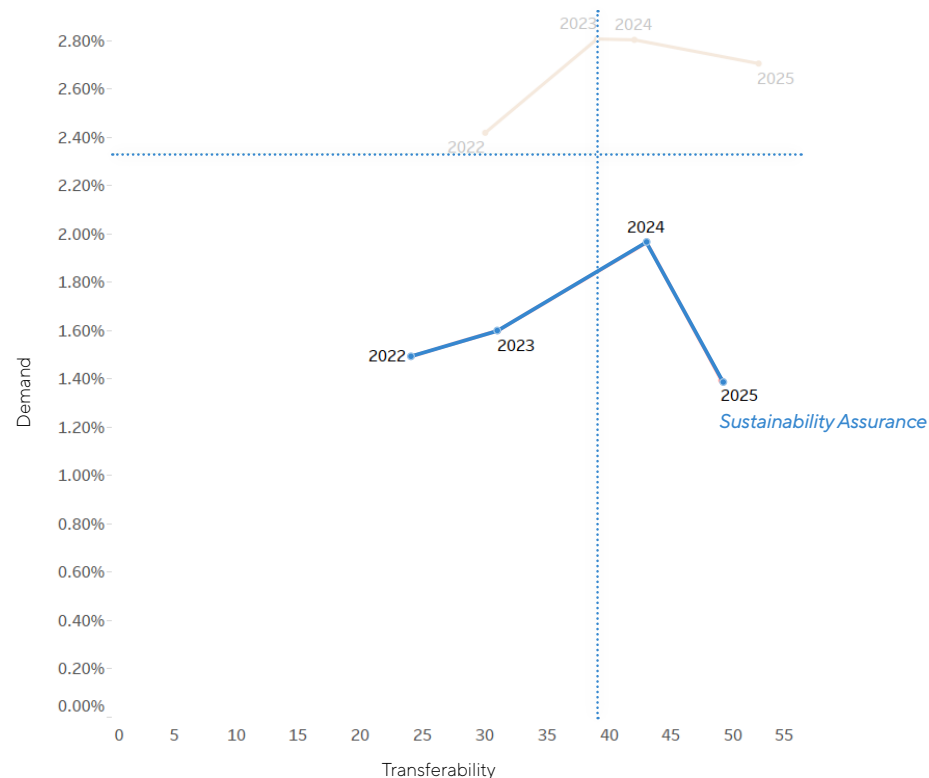
2 skills, or about 4% of 47 skills, crossed the 70th percentile mark between 2022 and 2023 in terms of transferability, attributable to an increase in skills application across job roles in the economy. Among them, *Sustainability Assurance* was selected for illustration.

This skill progressed from the Niche quadrant to the Transferable quadrant and is projected to remain so in 2025.

Job roles requiring this skill include premises and facilities maintenance managers, business consultants, and business and financial project managers.

Singapore's mandate for climate-related disclosures – starting from 2025 for listed firms and 2027 for large non-listed firms – underscores this shift. Organisations increasingly require *Sustainability Assurance* experts who can evaluate and verify the accuracy, relevance, and completeness of corporate sustainability reporting. This expertise is becoming crucial for companies aiming to comply with evolving reporting requirements and demonstrate their commitment to environmental goals, social responsibilities, and financial performance.

Movement of Priority Skills in the Green Economy related to Observation Two

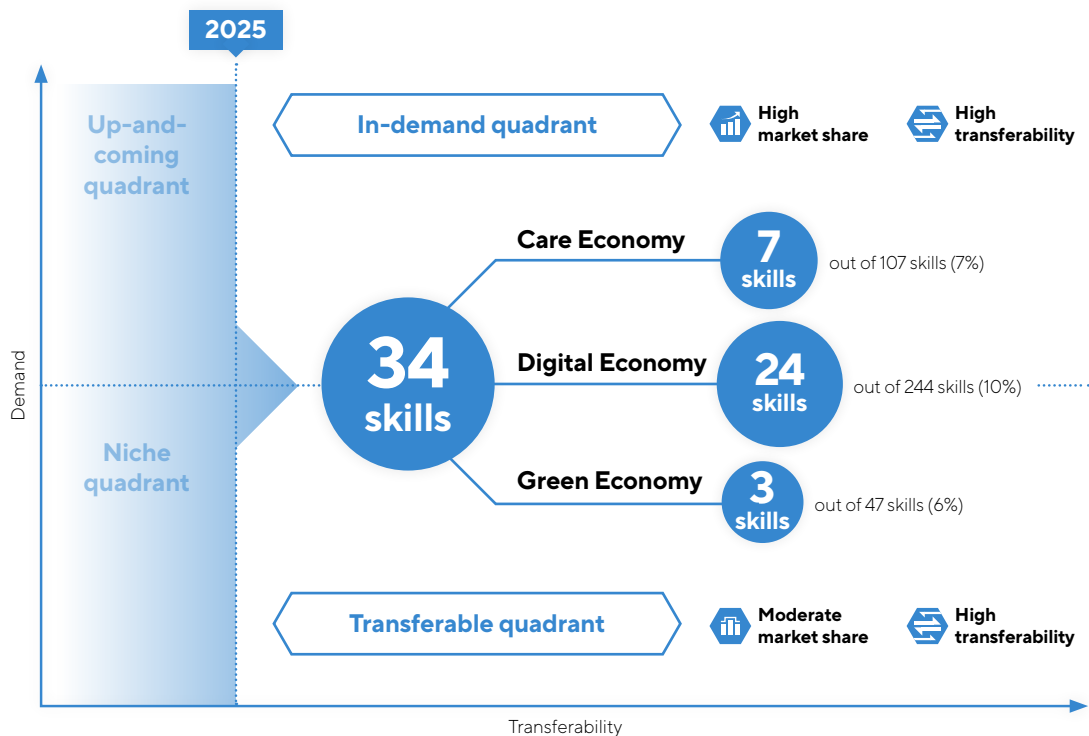




Observation Three: Skills forecasted to become highly transferable in 2025

34 skills are forecasted to become highly transferable in 2025.

A total of 34 skills, or about 9% of 398 skills across the three economies, are forecasted to cross the 70th percentile mark in terms of their transferability in 2025 after being in the same quadrant for the past three years. On the other hand, 309 skills, or about 78% of 398 skills, stayed in the same quadrant between 2022 and 2024, and are forecasted to remain.





Observation Three: Skills forecasted to become highly transferable in 2025

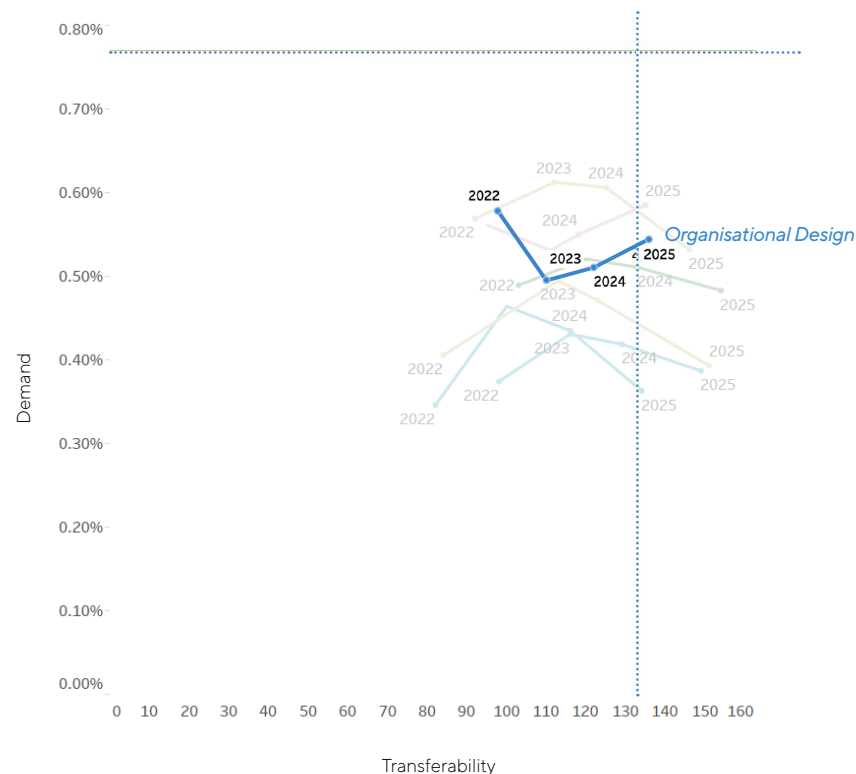
Within the Care Economy, 7 skills are expected to become highly transferable in 2025.

7 skills, or about 7% of 107 skills, were consistently in the same quadrant across the past three years. However, these skills are projected to change quadrants in 2025. Among them, *Organisational Design* was selected for illustration.

Between 2022 and 2024, *Organisational Design* remained in the Niche quadrant but is projected to progress into the Transferable quadrant.

Job roles requiring this skill include administration managers, HR managers, and software developers. This skill is increasingly important to support greater adoption of hybrid work models¹⁰ and transition toward agile and flexible organisational structures that decentralises the responsibility for organisational design from central departments to business units and line managers.

Movement of Priority Skills in the Care Economy related to Observation Three



¹⁰ The Straits Times, 2023.



Observation Three: Skills forecasted to become highly transferable in 2025

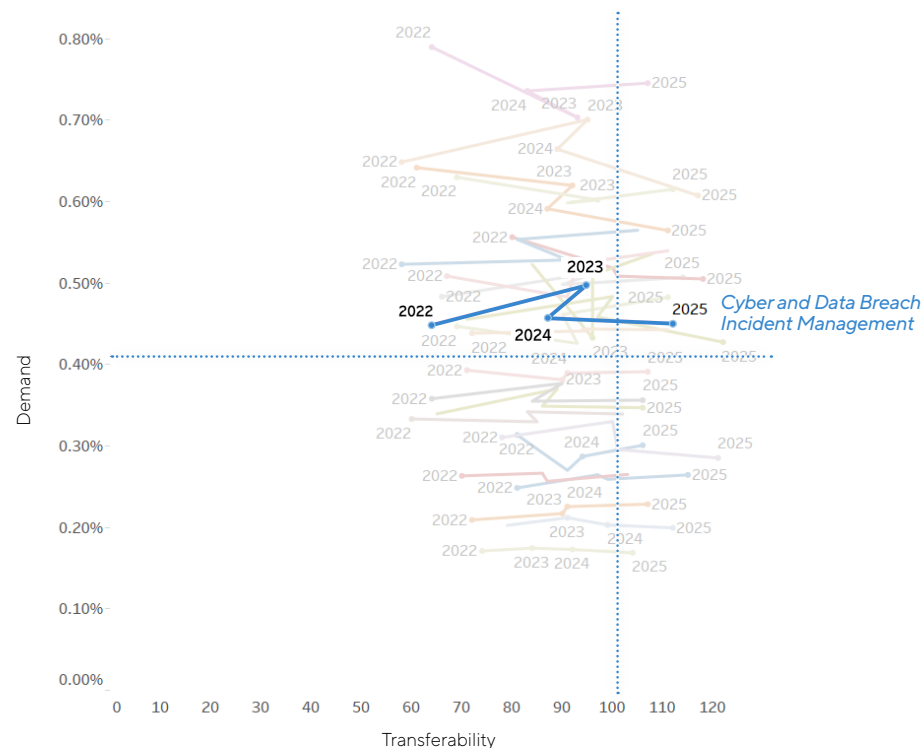
Within the Digital Economy, 24 skills are expected to become highly transferable in 2025.

24 skills, or about 10% of 244 skills, were consistently in the same quadrant across the past three years. However, these skills are projected to change quadrants in 2025. Among them, *Cyber and Data Breach Incident Management* was selected for illustration.

In 2024, *Cyber and Data Breach Incident Management* was in the Up-and-coming quadrant. In the coming year, it is projected to progress into the In-demand quadrant.

Job roles requiring this skill include IT support technicians, cyber risk specialists, and IT infrastructure specialists. This skill helps employees to detect breaches, trigger alerts, and resolve incidents. The number of data breaches continues to be on the rise and cyber threats are evolving with the help of large language models¹¹. Attacks can be carried out against cloud infrastructure, or use stolen credentials, to disguise their activities as legitimate¹². Organisations will need to improve on how these incidents are identified, reported, and dealt with effectively in the face of increasing threats. As cyber incidents proliferate, roles beyond IT security – such as web and mobile developers, financial analysts, and consultants – will also need this skill to help prevent or swiftly mitigate potential damage.

Movement of Priority Skills in the Digital Economy related to Observation Three



¹¹ World Economic Forum, 2024.

¹² CrowdStrike, 2024.



Observation Three: Skills forecasted to become highly transferable in 2025

Within the Green Economy, 3 skills are expected to become highly transferable in 2025.

3 skills, or about 6% of 47 skills, were consistently in the same quadrant across the past three years. However, these skills are projected to change quadrants in 2025. Among them, *Green Facilities Management* was selected for illustration.

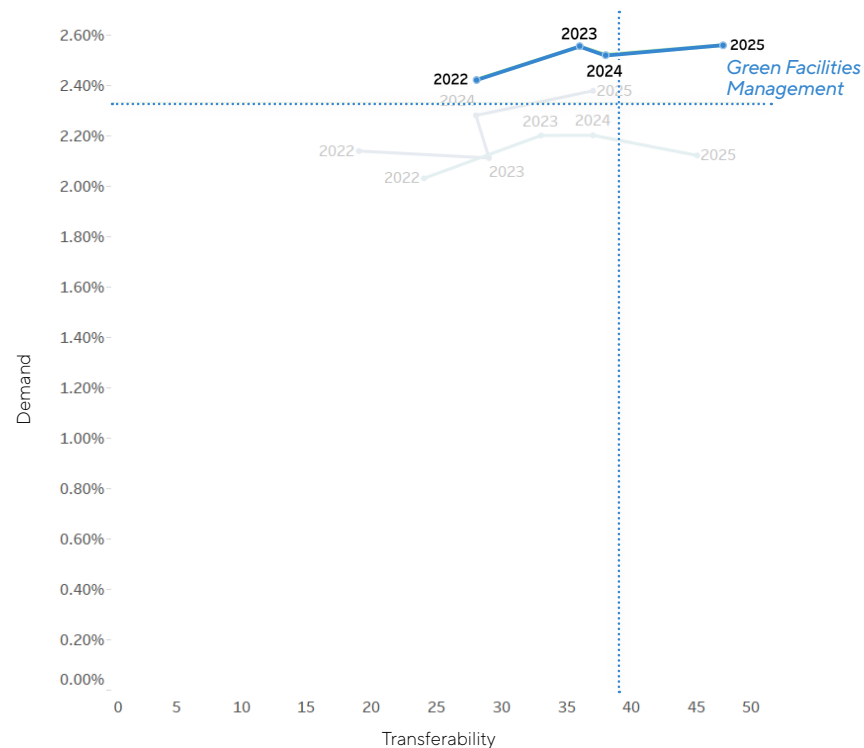
In 2024, *Green Facilities Management* was in the Up-and-coming quadrant. In the coming year, it is projected to progress into the In-demand quadrant.

Job roles requiring this skill include mechanical engineers, business and financial project managers, and restaurant managers.

The growth is driven by three megatrends:

- Global decarbonisation: Increased demand for sustainable buildings in Singapore and Asia Pacific^{13,14}, requiring this skill for eco-friendly infrastructure and efficient operations.
- Technological advancements: Improved viability of green buildings through reduced embodied carbon and enhanced efficiency¹⁵, using smart systems, IoT, and AI for energy monitoring.
- Recognised¹⁶ benefits: Wider acknowledgement of green buildings' advantages, including lower operational costs, higher property values, and improved employee productivity, further driving demand for this skill.

Movement of Priority Skills in the Green Economy related to Observation Three



¹³ EdgeProp, 2024.

¹⁴ Jones Lang LaSalle Incorporated (JLL) 2024.

¹⁵ The Business Times, 2024.

¹⁶ Swan & Maclaren, 2024.

Special Mention: Featuring 11 skills poised to capture greater demand and transferability in the future.



Care Economy

- *Ethical Conduct and Professional Integrity*



Digital Economy²²

- *Use of Generative AI Models*
- *Ethical and Governance Considerations in Generative AI*
- *Core Principles of Generative AI*
- *Prompt Engineering*
- *Generative AI Evaluation*
- *Generative AI Model Development*
- *Advanced Generative AI Technology*
- *Application Development of Generative AI Models*



Green Economy

- *Greenhouse Gas Accounting*
- *Carbon Footprint Management*

The Skills Matrix has been central in this chapter to showcase skill movement trends. This section moves away from data-driven insights to uncover 11 skills across three economies. While these skills did not have substantive local job posting data to exhibit prominent transferability and demand characteristics, they are poised to capture greater demand and transferability in the future.

In the Care Economy, *Ethical Conduct and Professional Integrity* is a skill needed to navigate new areas of ethical dilemmas amidst societal and technological advances¹⁷. With technology integration leading to more sharing of personal data, there is a greater need for professionals to exercise this skill to uphold the quality of care, prevent the misuse of client data, ensure client confidentiality, and maintain trust¹⁸. Job roles requiring this skill include human resource managers, nurses, and health services managers.

In the Digital Economy, inaccuracy, intellectual property infringement¹⁹, and low-quality output²⁰ are among the top risks of using GenAI. Broad-based skills that are needed to mitigate these risks are:

- *Use of Generative AI Models*
- *Ethical and Governance Considerations in Generative AI*
- *Core Principles of Generative AI*
- *Prompt Engineering*

AI users (enterprises and general workforce) who are equipped to use AI-powered products and services to increase productivity, and pursue better jobs and more impactful work²¹, will require

such skills to ensure responsible, transparent, and accountable usage of AI technologies.

As the area of GenAI continues to evolve rapidly, specialised GenAI skills will be used to apply more types of solutions regularly across a broad range of areas such as IT, Research, and Engineering. These specialised skills which develop and integrate AI models and processes are:

- *Generative AI Evaluation*
- *Generative AI Model Development*
- *Advanced Generative AI Technology*
- *Application Development of Generative AI Models*

Compared to the more general usage of GenAI skills, AI practitioners (tech workers) with the skillsets to create, implement, and deploy AI systems, models, and algorithms in organisations, at scale²³, will need these skillsets to lead GenAI research and deploy GenAI systems.

In the Green Economy, growing climate concerns²⁴, stricter regulations, and consumer demands and emphasis on sustainability will drive the importance of these skills. *Greenhouse Gas Accounting* quantifies emissions and *Carbon Footprint Management* uses this data to reduce environmental impact. Together, they enable organisations to measure emissions, develop strategies, comply with regulations, and meet stakeholder expectations²⁵. Professionals in management and compliance, particularly sustainability officers and analysts, will need these skills to adapt to a low-carbon economy.

¹⁷ Hadjistavropoulos, Thomas, American Psychological Association, 2023.

¹⁸ Deloitte, 2021.

¹⁹ McKinsey, 2024.

²⁰ Bain & Company, 2024.

²¹ Smart Nation Singapore, 2023.

²² Infocomm Media Development Authority, in partnership with SkillsFuture Singapore, and in consultation with industry stakeholders, has identified and developed a suite of GenAI skills.

²³ Smart Nation Singapore, 2023.

²⁴ World Economic Forum, 2024.

²⁵ Global Association of Risk Professionals, 2023.

CHAPTER 3

Creative Industries Skills Spotlight

Creative Industries are at the forefront of developing and adapting to emerging business trends. Technological advancements have opened new horizons, transforming how creative content is generated and experienced.

For instance, Changi Airport has used 3D modelling, simulation, and computer graphics to develop a 14-meter-tall LED waterfall titled 'Wonderfall' to create a magical atmosphere for visitors¹. On the other hand, the National Library Board's (NLB) StoryGen uses GenAI to create immersive multi-media experiences from text².

To understand the Creative Industries' current state, its influence, and its future trajectory in light of technological advancements, the following were studied:

1. The skills and tools required by Creative professionals
2. The growing demand for creative skills in non-creative sectors
3. The impact of advanced technologies like GenAI on creative work

The [Job Requirements Dashboard](#) accompanies some of the insights for this chapter. Please refer to the [Jobs-Skills Portal](#) for the dashboard.

Click [here](#) to watch videos showcasing our partners' perspectives.

¹ Changi Airport Group, 2023.

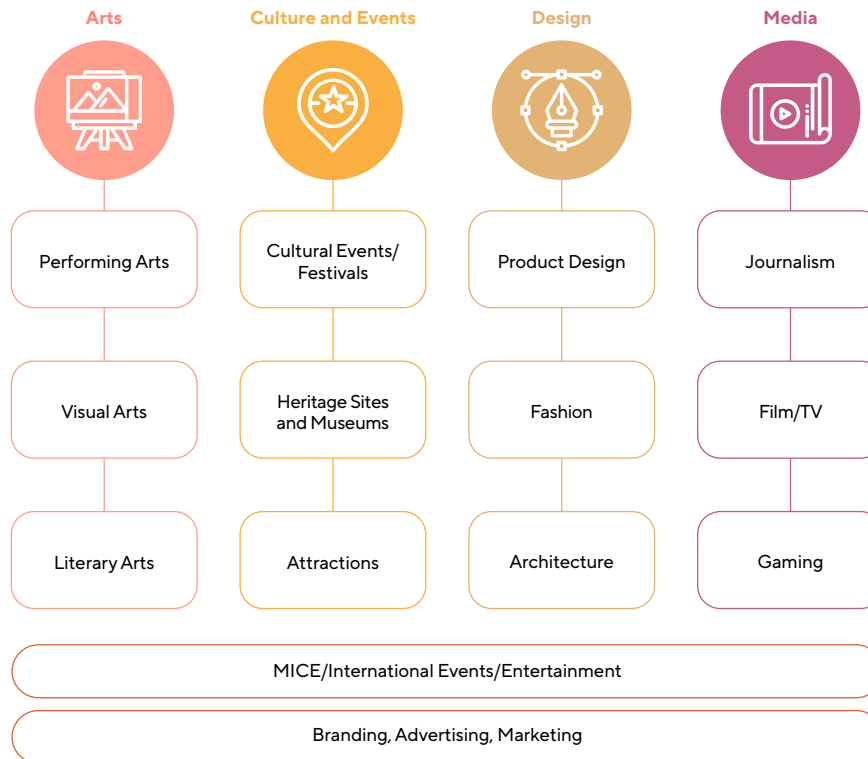
² National Library Board, 2024.

With technological advancements, Singapore's Creative Industries continue to evolve.

There are an estimated 150,000 Creative professionals in Singapore. Not all of them work in a creative sector. In fact, only a third of them work directly in a creative sector, while the rest work in a non-creative sector³.

Singapore ranked fifth in Global Innovation, partly contributed by the Creative Industries⁴.

Market Map of Singapore's Creative Industries



³ Data from LinkedIn Talent Insights.

⁴ World Intellectual Property Organisation, 2023.

For Creative professionals, the skills they chiefly need lie in the 'Business and Financial Management' and 'Organisational and People Management' clusters.

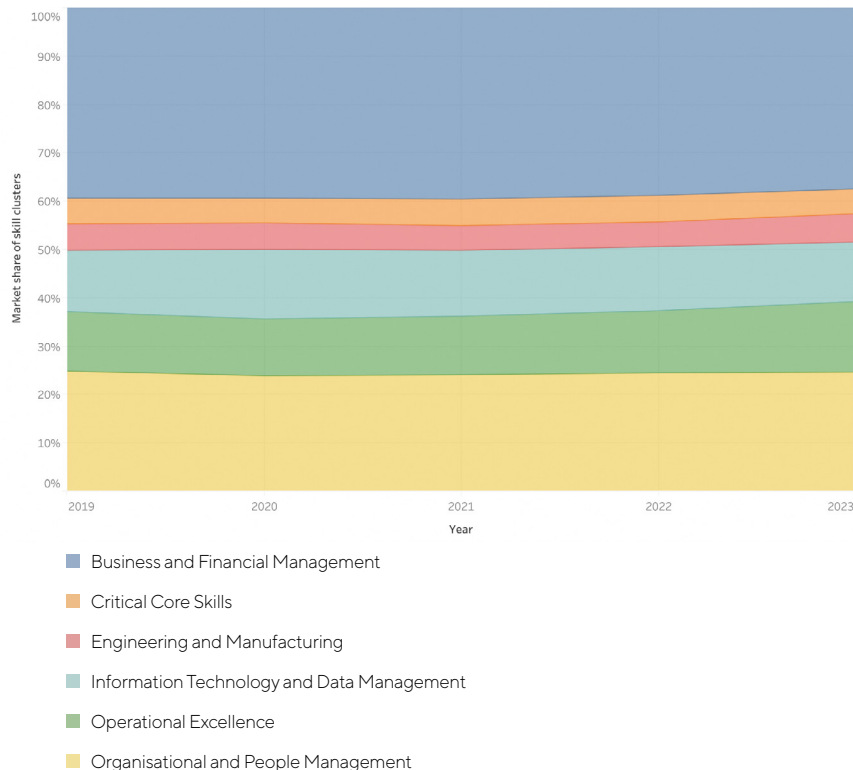
As Covid-19 resets many business norms, the skill demand of Creative professionals has witnessed shifts in the past five years.

From 2019 to 2023, demand for skills in 'Business and Financial Management' and 'Organisational and People Management' clusters remained stable, comprising more than half of the total skills required.

'Operational Excellence' skills increased since 2021, possibly due to the return of economic, cultural, and tourism related activities post-Covid⁵. The demand for skills such as *Meetings, Incentives, Conferences, and Exhibitions (MICE) Content and Experience Development and Delivery* has increased to ensure the smooth running of attractions, festivals, and large-scale events.

For skills in 'Information Technology and Data Management', 'Engineering and Manufacturing', and 'Critical Core Skills' clusters, the demand is relatively stable with slight fluctuations.

Skill Clusters of Creative Professionals



5 Singapore Tourism Board, 2022.

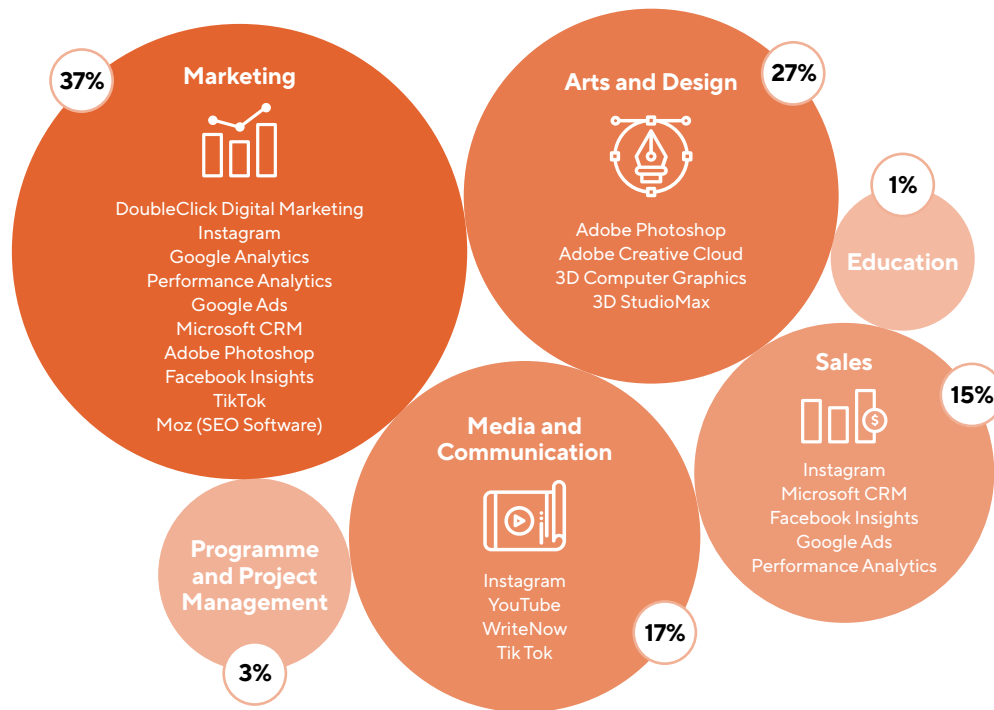
More than 3,000 Apps & Tools support the work of Creative professionals.

A wide range of digital tools are used by Creative professionals to support their work and amplify their creative voice. These include not just tools that aid in content creation, but also tools for sales and project management.

More than 3,000 Apps & Tools are needed by Creative professionals, largely to support work functions in marketing, arts and design, and media and communications.

A deeper look at the 100 most common Apps & Tools further revealed that 40% of them are AI-related.

Examples of Apps & Tools Required by Creative Professionals by Work Functions (2023)



AI-related Apps & Tools help non-Creative professionals to do creative work.

Creative tools are now widely available, allowing many people to do creative work, even if it's not their main job. As more of these tools are embedded with AI, existing and new tools will further expand user capabilities and provide new avenues for creative work across disciplines.

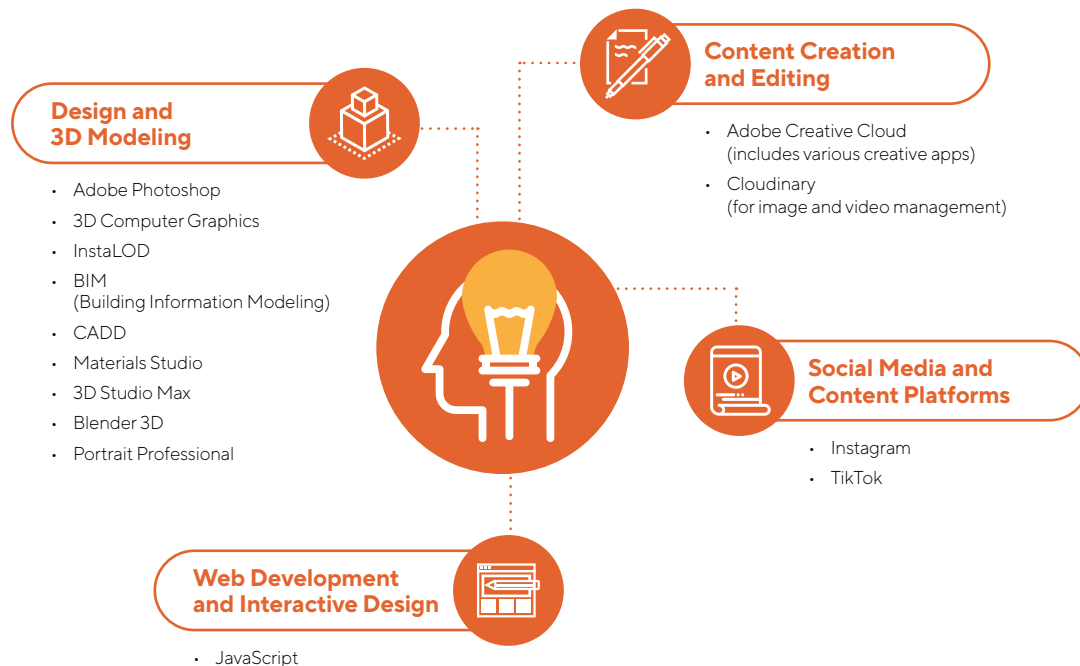
For instance:

- New tools help with various creative tasks:
 - In jewellery making, tools can turn 2D images into 3D models quickly⁶.
 - These tools also make realistic images, saving time and materials.
- Social media platforms like Instagram and TikTok help businesses:
 - Create engaging content easily
 - Showcase products and build brand awareness

With these tools, more professionals can now be creators in their work.

The availability and accessibility of creative tools represents a shift in how professionals both within and outside of the Creative Industries can take on the role of a creator at work.

Examples of Apps & Tools Required for Creative Work (2023)



7 out of 10 non-creative job roles need creative skills.

Creative skills are becoming more important in many jobs. A total of 40 creative skills experienced an increase in transferability since 2019. This means that more job roles are increasingly asking for creative skills.

7 in 10 non-Creative professionals require creative skills notably in IT, Engineering, Operations, Sales, and Business Development functions.

Examples of job roles needing creative skills:

Software developers: *Design Creation and Development* is needed to create user-centric applications with intuitive and visually appealing interfaces that enhance users' experience.

Restaurant managers and chefs: *Customer Acquisition Management* and *Brand Management* have gained importance to establishments that differentiate and manage their unique brand identity and continue to attract and grow their customer base.

This trend shows that having some creative skills can help people in many different careers.

Bubble Chart of Skills in The Creative Industries That Increased in Transferability from 2019 to 2023



A Survey Snapshot of Creative Professionals

Poll of Creative professionals:

Mixed sentiments towards the increased usage of GenAI tools, though many are using the tools to augment their creative work.

In a poll conducted from July to September 2024, 87 Creative professionals shared their sentiments on how GenAI Apps & Tools impacted their work.



Marketing ourselves as an all-human team might pay more dividends.〰



Would GenAI devalue the creative mind behind the work?〰

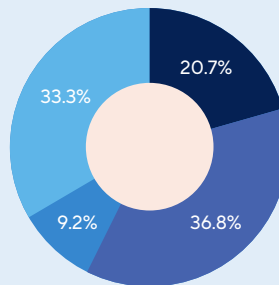


It's great for inspiration and thought-starters.〰



There may be an over reliance on GenAI.〰

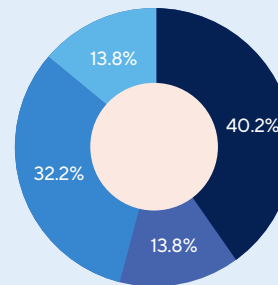
Have you used GenAI at work over the past 3 years?



- Yes, extensively
- Yes, occasionally
- No, but considering it
- No, not interested

More than **60%** had used or considered using GenAI in their work in the past three years.

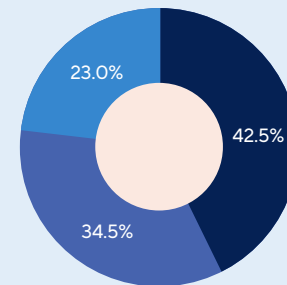
Overall, what is your view of GenAI's impact on your work?



- Scope of work is completely transformed
- Large impact
- Slight impact
- No impact at all

85% agreed that the GenAI tools have an impact on their work.

Overall, how do you feel about GenAI in your work?



- Positive
- Negative
- Unsure

40% felt positive towards the advancement of GenAI, slightly more than those who felt negative.
23% were unsure.

More than 50% of Creative professionals had used some form of GenAI tools in their work.

Creative professionals in Singapore are fast embracing available AI-related tools in various aspects of their business operations. In a response to a survey, the three most frequently mentioned GenAI tools are: ChatGPT, Firefly and Midjourney. (Please refer to the [list of Apps & Tools](#) in the Jobs-Skills Portal for their descriptions).

They acknowledged the benefits brought about by the tools, and also shared their concerns over ethical use, data ownership, Intellectual Property (IP) infringement, and output quality.

GenAI has created both excitement and new opportunities in the creative field. However, Creative professionals stressed that it is crucial to use these tools wisely. They emphasised that human creativity should guide the use of GenAI, rather than relying on it entirely.

When used effectively, GenAI can make routine tasks more efficient, and inspire fresh ideas.

The key is to view GenAI as a tool that enhances human creativity, not as a replacement for it. By combining human ingenuity with GenAI's capabilities, Creative professionals can achieve better results in their work.

Text and Language



ChatGPT



Gemini



Meta AI
LLaMA



Claude



PaLM 2



fireflies.ai

Collaborative and Project Management

FigJam

TeamGPT

Image Generation and Editing



DALL-E

stability ai



Midjourney



alpaca



Photoleap



Canva



Adobe Firefly



Pika AI



BRIA

Code and Development Assistance



Copilot



Microsoft
Azure



IBM



STORY-
BOARDS.ai

scriptreader.ai

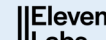


Colourlab AI

Film and Media Production



FILMUSTAGE



Eleven
Labs



runway



Beneficial impacts of GenAI tools in the Creative Industries include

- Inspiring creativity
- Streamlining administrative tasks
- Improving efficiency



Some **concerns** towards GenAI Creative professionals face include

- Ethical concerns
- Ownership of data
- Intellectual property
- Quality of data output and confidentiality

CHAPTER 4

Explore Pathways to In-Demand Jobs

(In partnership with Workforce Singapore)

The previous edition of the SDFE report examined how individuals can take charge of their career planning, and how certain jobs can be more attainable by adopting a multi-step approach to enhance career mobility, bringing individuals closer to their desired jobs.

In this edition, the chapter focuses on examining the interplay between five key factors individuals should consider when seeking career opportunities:

- ① **Wage¹** serves as the indicator of value that employers place in the job.
- ② **Demand for jobs** refers to the extent to which the job market is seeking to hire for that job role. Understanding the demand for certain jobs can help individuals identify areas where their skills and expertise are sought after, and what some potential career opportunities are.
- ③ **Growth of job roles** refers to the trend in the demand for the job over the past three years. Recognising job roles with positive growth trends can guide individuals towards job roles that have growth prospects and advancement opportunities.
- ④ **Skills similarity** refers to the amount of skills overlap between two jobs. It helps the individuals identify the relevance and transferability of one's existing skills to different job roles, either within the same sector or across sectors. Recognising similarities between current and desired skills can guide individuals on the effort and feasibility of making a career move.
- ⑤ **Historical transitions** across job roles refer to actual movements of workers from one job to the other. Understanding historical transitions can provide insights into the adaptability and resilience of certain career moves, helping individuals make informed decisions².

On top of these criteria, career development professionals also highlight that individuals can consider past experience, resources required in terms of time and effort, and affordability when making a career move, other than their own values, interest, personality, and skills (VIPS).

The five factors are incorporated into a Skills and Job Mobility Dashboard which can be used to explore career pathways that can lead to growth opportunities.

The [Skills and Job Mobility Dashboard](#) was used to generate the insights for this chapter. Please refer to the [Jobs-Skills Portal](#) for the dashboard.

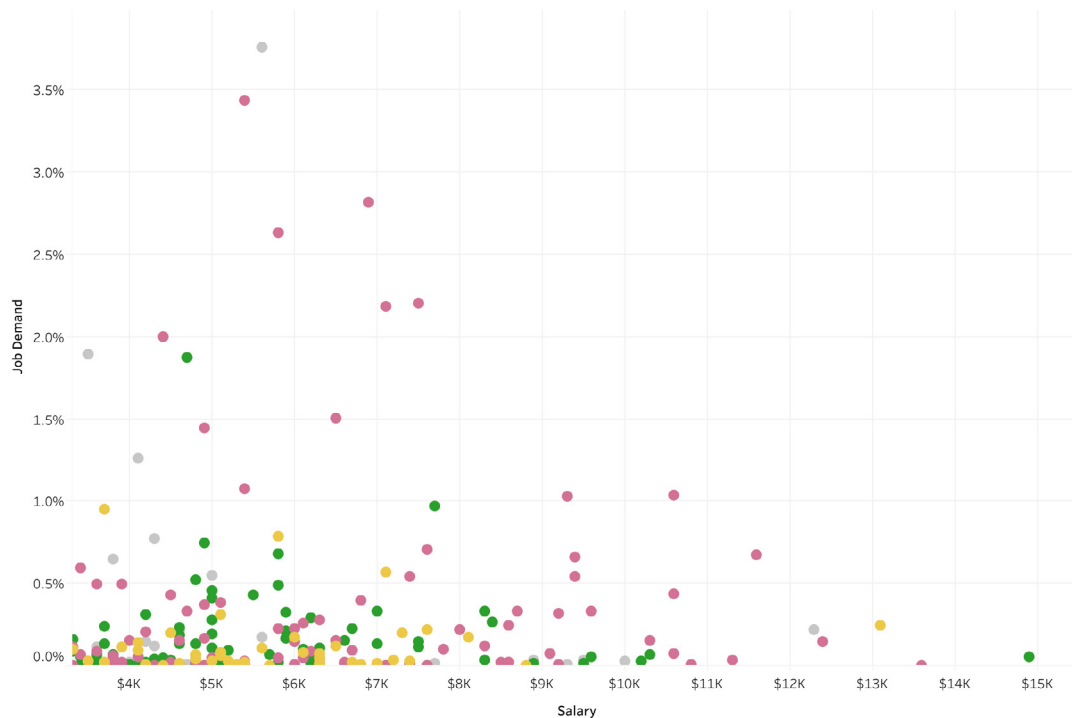
146 in-demand job roles with good growth and potential for career mobility were identified based on 2023 job posting data.

This analysis is limited by the availability of wage and demand data. Out of a total of 1,002 job roles, wage and demand data were obtained for 342 job roles.

Of these 342 job roles, 146 good job roles termed as 'job roles with good growth and potential for career mobility' was shortlisted by applying three criteria:

1. Jobs that offer a wage of S\$3,300 and above
2. Jobs that are in demand by employers
3. Jobs that are growing in demand

Distribution of Job Roles by Salary and Demand in 2023



Economy

- Care
- Digital
- Green/I4.0
- Other

Sample of Job Roles with Good Growth and Potential for Career Mobility

Job roles	Median Gross Wage	Job Demand (No. of job postings)	Job demand (Market share)	Change in job demand
Administration manager ³	S\$5,600	24,289	3.76%	15.95%
Marketing manager ⁴	S\$6,900	18,192	2.82%	0.17%
Logistics planner ⁵	S\$4,700	12,072	1.87%	9.74%
Customer service manager ⁶	S\$8,100	1,123	0.17%	5.56%

1 In this chapter, all mentions of wages/salary refer to Median Gross Wage and reference the Occupational Wage Survey, provided by the Manpower Research & Statistics Department from the Ministry of Manpower (MOM) and are rounded to the nearest hundred

2 'In this chapter, all mentions of historical transitions reference the Occupational Employment Dataset provided by the Manpower Research & Statistics Department from the Ministry of Manpower

3 Based on Singapore Standard Occupation Classification (SSOC) code 12112: 'Administration manager'.

4 Based on Singapore Standard Occupation Classification (SSOC) code 12222: 'Marketing manager'.

5 Based on Singapore Standard Occupation Classification (SSOC) code 33461: 'Logistics/production planner'.

6 Based on Singapore Standard Occupation Classification (SSOC) code 12241: 'Customer service manager'.

Using the Skills and Job Mobility Dashboard with a Built-in Five-Factor Criteria to Analyse Job Transitions

The 146 job roles with good growth and potential for career mobility take centre stage in this Skills and Job Mobility Dashboard. A range of career transitions is possible, both into these roles, as well as eventually out to another job role.

The purpose of the dashboard is primarily for exploration and analysis only. It is recommended to have conversations with Career Coaches, Skills Ambassadors, or supervisors to further the exploration.

There are multiple approaches that individuals can undertake to fulfil their career moves. Job transitions may sometimes require a decrease in salary, depending on the needs of an individual. This situation can arise from various factors such as work-life balance, relocation, personal growth to gain experience in certain fields, job market conditions, and health considerations. It is important to note that while a salary decrease may seem disadvantageous, it can sometimes lead to long-term benefits such as improved job satisfaction, better work-life balance, or enhanced career prospects.

Learn to use the Skills and Job Mobility Dashboard, accessible on the Jobs-Skills Portal.

1

Identify your current job role in the 'Origin Job Role' drop-down list. Choose the job role which is the closest match to yours.

2

Select a job role that you are interested in exploring in 'First Destination Job Role'.

3

The fields in the 'Origin Job Role' to 'First Destination Job Role' will be populated based on backend algorithms. The job transitions are identified based on the job roles with the highest skills similarity score. These job roles are then shortlisted and appear in the drop-down list. The insights will show:

- Skills similarity between 'Origin Job Role' and 'First Destination Job Role'
- Historical Transition in the past two years
- Difference in wages (if any, positive or negative wage gain)
- Skills required for transition and corresponding courses if reskilling is required

4

Select 'Second Destination Job Role' if you are interested in knowing the possible move from 'First Destination Job Role'.

5

Similar to step three, the fields in the 'First Destination Job Role' to 'Second Destination Job Role' will be populated.

The dashboard is divided into five numbered sections, each with a title, a dropdown menu, and a table of metrics. Arrows indicate a sequential flow from step 1 to step 5.

1 Origin Job Role
 (Multiple values) [dropdown]
 Salary: Null
 Job Demand: Null
 Change in Job Demand: Null
 Economy: Null
 Select at most 1 job role

2 First Destination Job Role
 (Multiple values) [dropdown]
 Salary: Null
 Job Demand: Null
 Change in Job Demand: Null
 Economy: Null
 Select at most 1 job role

3 Origin to First Job Role
 Select at most 1 origin job role and 1 second job role.
 Skills Similarity: Null
 Historical Transitions: Null
 Difference in Salary: Null
 Skills required for transition: Null
 Course Title: Null

4 Second Destination Job Role
 (Multiple values) [dropdown]
 Salary: Null
 Job Demand: Null
 Change in Job Demand: Null
 Economy: Null
 Select at most 1 job role

5 First Job Role to Second Job Role
 Select at most 1 origin job role and 1 second job role.
 Skills Similarity: Null
 Historical Transitions: Null
 Difference in Salary: Null
 Skills required for transition: Null
 Course Title: Null

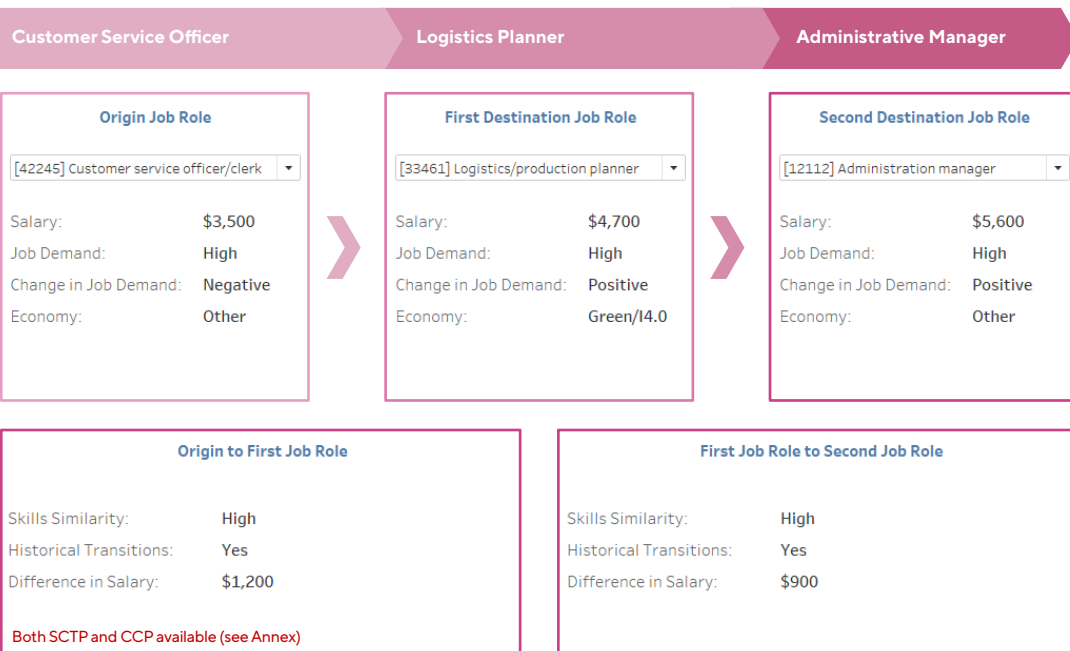
Example One: Moving across sectors where skills are similar between jobs

This career path demonstrates a strategic progression with increasing responsibility and wage potential. High skills similarity between these roles may facilitate smooth career advancement, showcasing the interconnected nature of these positions. Real-life career transitions have been observed among these three identified job roles.

Customer service officers⁷ build essential communication and problem-solving skills, which will then be required by logistics planners⁸. The additional skills required may include operational efficiency and coordination. Two to three years of work experience will typically be required as well.

One can enrol in a SkillsFuture Career Transition Programme (SCTP) for Advanced Certificate in Logistics and Supply Chain Management, or Career Conversion Programme (CCP) for Supply Chain and Logistics Professionals to prepare for the transition to logistics planner.

The next step to administrative manager⁹ – often achievable after three to four years as a logistics planner – capitalises on the combined customer-centric and operational expertise. Key skills required by an administrative manager include *Business Presentation Delivery*, *Change Management*, and *Stakeholder Management*.



⁷ Based on Singapore Standard Occupation Classification (SSOC) code 42245: 'Customer service officer/clerk'.

⁸ Based on Singapore Standard Occupation Classification (SSOC) code 33461: 'Logistics/production planner'.

⁹ Based on Singapore Standard Occupation Classification (SSOC) code 12112: 'Administration manager'.

Example Two: Developing technical skills to gain entry to in-demand jobs in today's digital age.

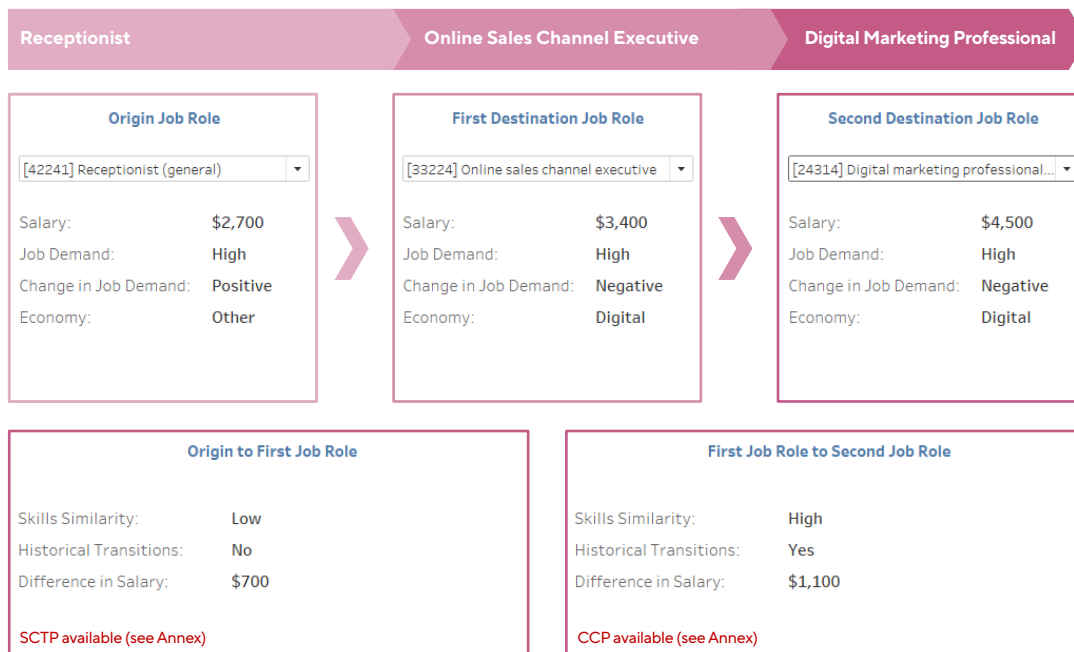
This career path demonstrates a strategic progression leveraging transferable skills and adapting to the digital economy.

Starting as a receptionist¹⁰ builds essential communication and organisational skills, which transition well into an online sales channel executive¹¹ role where digital communication and customer engagement become key, typically achievable after one to two years of work experience.

Despite low skills similarity and no historical job transitions between receptionists and online sales channel executives, one can consider enrolling in the SCTP for Omnichannel Specialist to prepare for the job transition.

The next possible step to digital marketing professional¹²—often achievable after two to three years of work experience—capitalises on the combined customer interaction expertise and digital platform knowledge gained in previous roles. The CCP for Digital Sales and Marketing Professionals aids this transition.

This three- to five-year pathway demonstrates progression from traditional to specialised digital marketing skills, offering increasing responsibilities and wage potential.



¹⁰ Based on Singapore Standard Occupation Classification (SSOC) code 42241: 'Receptionist (general)'.

¹¹ Based on Singapore Standard Occupation Classification (SSOC) code 33224: 'Online sales channel executive'.

¹² Based on Singapore Standard Occupation Classification (SSOC) code 24314: 'Digital marketing professional (e.g. online, social media, e-commerce marketing professional)'.

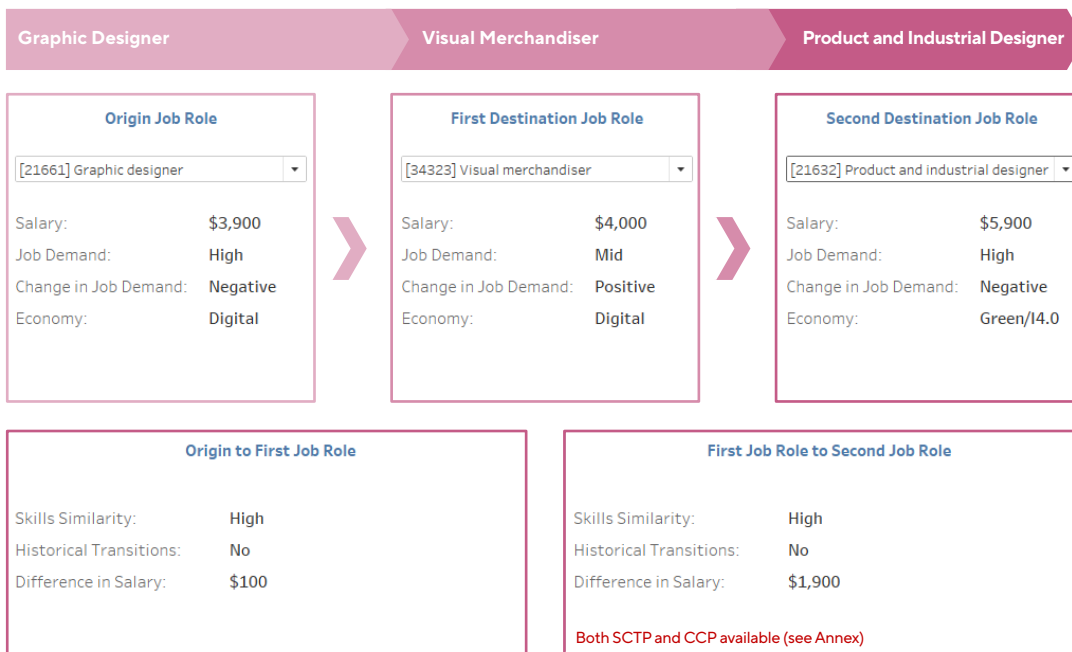
Example Three: Mastering current skills and deepening work experience in specialised work areas can open up career options.

This design-centric career pathway demonstrates how professionals can leverage their creative skills to transition across related fields.

Starting as a graphic designer¹³ builds a strong foundation in visual communication and digital tools, which translates well for a visual merchandiser¹⁴ where spatial design and brand representation become important. Key skills for visual merchandisers include *Brand Management* and *Digital and Physical Prototyping*.

The next possible step to product and industrial designer¹⁵ capitalises on visual communication skills, expanding into physical product creation. While historical data on these specific transitions may be limited, the high skills similarity between these roles suggests potential for successful career shifts. The SCTP and CCP are available to support the transition to the product and industrial designer role.

This transition typically takes five to eight years in total, offering increasing responsibilities and creative scope as one progresses from digital design to retail environment to product innovation.



¹³ Based on Singapore Standard Occupation Classification (SSOC) code 21661: 'Graphic designer'.

¹⁴ Based on Singapore Standard Occupation Classification (SSOC) code 34323: 'Visual merchandiser'.

¹⁵ Based on Singapore Standard Occupation Classification (SSOC) code 21632: 'Product and industrial designer'.

Call to Action



66
Plan and act towards fulfilling careers, with your desired wage growth and career mobility. 99

Tracy LEE

Assistant Chief Executive (Planning),
 Workforce Singapore



To make the best of longer working lives due to increased longevity, one could plan more purposeful career transitions, over the different stages of one's career¹⁶. While some career changes may be driven by one's immediate financial goals, a purposeful career transition is facilitated by proactive planning and consideration of one's values, interest, personality, and skills (VIPS), and purpose for a fulfilling career¹⁷.

Having a career plan also allows one to consider a longer-term view, where a new role might not offer an immediate salary increase but sets the stage for future career changes and growth.

To plan one's career in Singapore, there are self-help digital tools and resources available, such as CareersFinder and Workipedia by MyCareersFuture, as well as personalised career guidance provided by career coaches.

Maurice Ng, a Senior Career Coach at WSG, recalled a successful career transition by his client, a software engineer, whose contract was not renewed. He worked with his client to align career goals with his VIPS to shortlist vacancies that resonated with his career goals, and his client took courses in AI to close the skills gap between his previous and new job. Through tailored resume-crafting and intensive interview preparation, Maurice guided his client into a new role in Smart Technology, leading to a 20% wage increase and pivoting into an emerging role with longer-term career growth opportunities.

Take a well-planned leap, not into the unknown, but into purposeful career transitions to unlock new opportunities, get closer to your wage aspirations, and open the way to long-term professional growth.

¹⁶ Organisation for Economic Co-operation and Development, 2024.

¹⁷ Harvard Business Review, 2023.

Research Insights



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Flex and flexibility, finding yourself, virtual fitting rooms, and embodied cognition. These are new tools¹⁸ designed for successful multiple job transitions, termed Open Loop Careers in a recent research that we completed. The study emphasised that individuals must flex (i.e. demonstrate) their ability to perform in new jobs or roles, instead of claiming they can or have the qualifications. Workers also need to develop flexible work identities, identifying with their next position during career transitions.

Both flex and flexibility are challenging. To address this, we designed localised tools to engender earlier change and smoother transitions.

To exercise flexibility, we designed tools to find yourself:

- Career Quest Cards¹⁹: A gamified, fun and tactile self-discovery/assessment of trade-offs to clarify priorities
- Career Journey Maps: A tactile, visual, active reflection similar to reminiscence therapy and functional medicine

To exercise flexing, we built virtual fitting rooms that include:

- Skills Swap: A role-playing game to expand work options to the unfamiliar or unusual, yet plausible.

- What's Next GPT: Uses GenAI to map creative options to the whole-of-person i.e. work, life, and wellbeing

These tools engage multiple senses to engender earlier change, because when we “set the body in motion, the brain follows”²⁰. They adopt the cognitive science view of “embodied cognition” where brain, body, and environment are engaged in thinking and decision-making, unlike the conventional focus on the brain as a computer where change happens by “installing instructions”.

These tools also enhance other resources such as the Employability Self-Assessment for Older Adults, which reveals areas where flex and flexibility in work are enhanced.

Moreover, when stacked in sequence – where the output of one tool becomes the input for another – individuals feel a sense of progress, further engendering change. This approach increases the likelihood of successfully making multiple transitions.

Importantly, these new tools complement, recognise, and leverage existing skills like communication, collaboration, and problem-solving that are crucial when making job transitions. Together with these foundational competencies, the combinations provide a strong basis for adapting to new roles and industries, giving individuals a significant advantage in their career shifts.

¹⁸ Singapore University for Technology and Design, 2024.

¹⁹ Kickstarter, 2024.

²⁰ Carmen Simon, 2024.

Profile Story One

Selene Diong

Head of Enrolment and Marketing
at an Education Group

From Classroom to Marketing

Having spent years in education as an English teacher, Ms Selene Diong brings a unique perspective when she switched to take up a marketing role at a private education provider. Her teaching background allowed her to shape marketing strategies that are not only impactful but also deeply authentic, resonating with the community her organisation serves.

Acquiring Digital Skills for Career Transition

After spending much of her career in the classroom, she saw this as an opportunity to develop skills outside of teaching, which would allow her to grow professionally in new directions. This role also gives her the freedom to learn something highly transferable—digital marketing skills that can be applied across industries and future opportunities and resonates with her long-term goals. “It’s important that we market what we actually deliver in the classroom. My background as an educator gives me insights that help avoid a mismatch between what we promise and what we deliver”, shared Selene. As the Head of Enrolment and Marketing, she oversees the strategic direction for student enrolment and all marketing

efforts, ensuring effective communication of the value of their programmes to both students and parents.

Navigating the Learning Curve through a SkillsFuture Programme

Selene’s learning journey has been a mix of self-discovery and continuous learning. She realised that she needed a more structured approach to learning, which is when she signed up for the SkillsFuture Career Transition Programme (SCTP) Advanced Professional Certificate in Digital Marketing at NTU. She now approaches her role with more confidence, knowing that the skills she acquired are helping her contribute to both her personal and organisational goals.

Lifelong Learning is the key to Career Transition

Selene’s mantra is to embrace every opportunity to learn and grow. Lifelong learning is an investment that pays off both personally and professionally, and it is a reminder that every new chapter in life is a chance to reinvent oneself. The ability to adapt, embrace change, and acquire new skills is what will keep her competitive and fulfilled in her professional life.

Profile Story Two

Paramasivam S/O Rajagopal

Operations Executive

at an Integrated Facilities Management Company

Translating Military Precision into Civilian Operations

After 40 years in the Singapore Armed Forces (SAF), Mr Paramasivam S/O Rajagopal transitioned to facilities management at 55. "Retirement felt too early," he says. Now, as an Operations Executive, he oversees logistics, manages soft services like janitorial teams, and serves as a safety officer, applying his decision-making and problem-solving skills honed in the military.

Navigating Independent Decision-Making

The shift from a structured military environment to the corporate world was challenging. "In the military, we follow top-down instructions, but here, I have to make independent decisions, which keeps me engaged," Paramasivam explains. To adapt, he completed the Career Conversion Programme (CCP) for Professional Executives at NTUC LearningHub, gaining critical skills in *Equipment Maintenance*, *Project Coordination*, and *Problem Solving* to manage his role effectively.

Embracing Technology for Efficiency

Adapting to industry changes also required embracing technology. His team adopted digital apps to streamline logistics and maintenance, reducing paper use and supporting sustainability. This shift has improved operational efficiency and aligned with modern environmental standards.

Work-Life Balance and Continuous Learning

Paramasivam values work-life balance alongside technical expertise. "In my younger days, work was about securing a future. Now, it's about spending time with family and maintaining mental well-being," he shares. As Singapore advances towards digitalisation, Paramasivam combines his technical skills with adaptability. "It's essential to keep upgrading our skills to stay relevant," he says, embracing professional growth and personal balance.

Profile Story Three

Chris Tan

Assistant Community Manager
at a Designer Boutique Hotel

A Journey of Rediscovering Passion in Hospitality

Mr Chris Tan's path to his current role has been anything but conventional. After working on a kiwi orchard in New Zealand during a working holiday, he returned to his true passion—hospitality. "Hospitality has always drawn me in," he reflects on his studies at the Singapore Institute of Technology. Now, as Assistant Community Manager, he thrives in a role that combines community building with guest experience management.

Crafting Memorable Moments Beyond the Desk

"I can't sit behind a desk all day," Chris shares. He manages VIP guests, organises events, and ensures memorable stays by adding personal touches like nature walks and cooking classes. With 30% of guests staying long-term, Chris builds lasting relationships, which is a key aspect of his role.

From Picking Fruit to Picking Solutions

The transition from outdoor work to hospitality was made easier by the on-the-job training of his Career Conversion Programme for Tourism Professionals at the hotel he was working at and the supportive culture set by management. He mastered the hotel's cloud-based property management system and developed his prowess in *Event Planning and Management*. Chris also draws heavily upon Critical Core Skills like *Communication*, *Customer Orientation* and *Problem Solving* in his daily work. "No two complaints are the same, so you need to think on your feet," he says.

Turning Challenges into Opportunities

Chris thrives in the dynamic nature of his job, where managing guest expectations is a challenge he enjoys. His success stems from blending technical skills with interpersonal abilities, always learning and adapting. "Sometimes, you have to step out of your comfort zone," he says, embracing growth in hospitality.

Resources to Support Skills and Career Development

SSG and WSG offer a range of resources, tools, programmes, and initiatives to help you explore career opportunities, identify and acquire the necessary skills to increase employability, improve job performance, and adapt to job content changes.

Here are some specially curated resources to build a viable, long-term professional development career and skills strategy.

Jobs-Skills Portal

A one stop platform for all jobs-skills data needs. This includes:

SkillsFuture Jobs-Skills Insights

Publications that spotlight high-tempo jobs and skills changes within specific sectors or economies.

Singapore Skills Frameworks

Provides key sector information, occupations/job roles, and the required existing and emerging skills.

Careers & Skills Passport

Check out Careers & Skills Passport, a national digital repository containing Government-verified skills, employment, qualification and certification records.

<https://www.myskillsfuture.gov.sg/csp>

Skills and Training Advisory (STA) Services

Register 1-2-1 guidance from SSG Skills Ambassadors to discover the skills and training needed to achieve personal career goals.

<https://www.myskillsfuture.gov.sg/content/portal/en/career-resources/career-resources/education-career-personal-development/skills-training-advice.html>

SkillsFuture Advice

Sign up for SkillsFuture Advice, a 45-to-60-minute workshop session that shares more about SkillsFuture programmes and resources for career planning and upgrading needs.

<https://www.myskillsfuture.gov.sg/content/portal/en/career-resources/career-resources/education-career-personal-development/skillsfuture-advice.html>

Career Advisory and Coaching

Register for guidance from WSG Career Coaches with personalised career advisory and coaching services.

<https://www.wsg.gov.sg/home/individuals/career-matching-guidance>

Resources to Support Skills and Career Development

SkillsFuture Work-Study Programmes

Includes facilitated learning, on-the-job training, and work-based projects, to allow deepening of skill sets at the workplace.

www.skillsfuture.gov.sg/workstudy

SkillsFuture Career Transition Programme (SCTP)

Explore opportunities with this train-and-place programme which supports mid-career individuals in acquiring industry-relevant skills to improve employability and pivot to new sectors or job roles.

<https://www.myskillsfuture.gov.sg/content/portal/en/career-resources/career-resources/education-career-personal-development/skillsfuture-career-transition-programme.html>

WSG Career Conversion Programmes

Targeted at mid-career individuals, to undergo skills conversion and switch to new jobs or sectors.

<https://www.wsg.gov.sg/home/individuals/attachment-placement-programmes/career-conversion-programmes-for-individuals>

Skills Profiling Tools

Benchmark your employees' competencies to discover their strengths and skills gaps and identify suitable training through this tool (complimentary for SMEs)

<https://skillsfuture.gobusiness.gov.sg/support-and-programmes/skills-profiling-for-smes>

SkillsFuture Queen Bee Networks

Join a Queen Bee network to receive hands-on support to strengthen your business strategy.

<https://skillsfuture.gobusiness.gov.sg/support-and-programmes/skillsfuture-queen-bee-networks>

National Centre of Excellence for Workplace (NACE)

Receive assistance from experts to develop systems and processes to build your workplace learning capabilities. Fast track the skills certifications for your employees under the Workplace Skills Recognition Programme.

<https://skillsfuture.gobusiness.gov.sg/support-and-programmes/national-centre-of-excellence-for-workplace-learning-nace>

Resources to Support Skills and Career Development

Courses on MySkillsFuture Portal

Explore a wide range of SSG-supported courses to upskill, reskill, and stay competitive in today's job market.
<https://www.myskillsfuture.gov.sg/content/portal/en/training-exchange/course-landing.html>

Curated Courses for Critical Core Skills

A set of 16 transferable soft skills and competencies that enable employees to remain nimble and agile to support business transformation.
<https://go.gov.sg/ccs-courselist>

Critical Core Skills Profiling Tool

An assessment tool to determine your strengths and gaps in critical core skills.
<https://www.ial.edu.sg/resources/ccs>

SSG-supported Courses for Employers

Full suite of SSG-supported courses to support employers in their workforce development needs.
<https://skillsfuture.gobusiness.gov.sg/course-directory/search>

Curated Courses for Emerging and In-demand Skills for Individuals

A curated list of short, industry-relevant training programmes that focuses on emerging and in-demand skills in our economic growth pillars.
<https://www.myskillsfuture.gov.sg/content/portal/en/career-resources/career-resources/education-career-personal-development/skillsfuture-series.html>

Curated Courses for Emerging and In-demand Skills for Employers

A curated list of short, industry-relevant training programmes that focuses on emerging and in-demand skills in our economic growth pillars.
<https://skillsfuture.gobusiness.gov.sg/support-and-programmes/workplace-learning/skillsfuture-series>

SkillsFuture Enterprise Credit

A one-off \$10,000 credit for eligible employers to further reduce out-of-pocket expenses on qualifying costs for supportable upskilling and productivity initiatives as part of your business strategy.
<https://skillsfuture.gobusiness.gov.sg/support-and-programmes/funding/skillsfuture-enterprise-credit-sfec>

Exemplary Users of SkillsFuture Singapore's Jobs-Skills Resources

Many partners have innovatively used SSG's job-skills resources in the course of their work. Four of them are highlighted here. Watch these videos on the [Jobs-Skills Portal](#) for their exemplary use of the jobs-skills resources. May their stories inspire you to make full use of the various jobs-skills resources.

Singapore Management University

Using Skills Framework data to support staff in shaping their growth

Bone Marrow Donor Programme

Leveraging on data from Skills Framework and SDFE report to support skills development for staff

Singapore Pools Academy

Leveraging data from Skills Framework and SDFE report to align staff development to emerging trends

Avid Adventures

Adapting from Skills Framework and SDFE report insights to build digital career simulation game to support individuals

Methodology

Analysis methodologies applied in Chapter 1: Changes in Requirements of Job Roles

A. Tagging job posting data (2019-2023) to Singapore Standard Occupational Classification (SSOC)

Job posting data was tagged to the SSOC 2020, and at SSOC level 5, which focuses on the job role level. This granular classification allows for a clear distinction between different job roles, providing deeper insights into job demand and specific skill requirements.

B. Tagging job posting data (2019-2023) to a more refined skill clustering

In the previous edition of the SDFE report, skills extracted from job posting data were classified into the Singapore Skills Taxonomy (SST) level 1 which has 10 clusters, with the inclusion of Apps & Tools to make a total of 11 clusters.

In this edition, a data-driven approach was taken to refine the clustering of skills to six clusters based on broad business functions and critical core skills, reflecting the skill requirements needed across job roles in the Singapore economy.

The six **skill clusters** are: Business and Financial Management, Critical Core Skills, Engineering and Manufacturing, Information Technology and Data Management, Operational Excellence, and Organisational and People Management.

C. Tagging Applications and Tools to job posting data (2019-2023)

To keep pace with evolving needs, SSG has expanded its database to encompass over 5,000 unique **Apps & Tools**. In this report, Apps & Tools were mapped to work functions via job roles. Since each job role was mapped to one work function and each item in the list of Apps & Tools can be mapped to more than one job role, each item in the list of Apps & Tools can be mapped to more than one work function.

To derive the proportion of Apps & Tools used by respective work functions, the total count of Apps & Tools belonging to the work function was divided by the total count of Apps & Tools in 2023.

The 24 Apps & Tools clusters based on work functions are: Accounting, Administrative, Arts and Design, Business Development, Community and Social Services, Consulting, Customer Success and Support, Education, Engineering, Finance, Healthcare Services, Human Resources, Information Technology, Legal, Marketing, Media and Communication, Military and Protective Services, Operations, Program and Project Management, Purchasing, Quality Assurance, Real Estate, Research, and Sales.

For an additional layer of analysis, **Apps & Tools** were also categorised as **AI-related** if they contributed to the creation, development, or deployment of AI, leverage AI capabilities, or are associated with AI in general.

Analysis methodologies applied in Chapter 1: Changes in Requirements of Job Roles

D. Performing data analysis for Skills

In a given year, to derive the **demand of a skill cluster**, the total count of skills within each skill cluster was divided by the total count of skills across all 6 skill clusters based on job postings in a year and was presented as a percentage. This methodology can be applied at two levels: the job role level, which reveals the relative importance of different skill clusters within specific job roles, and the economy level, which illustrates the distribution of skill cluster demand across the entire Singapore economy.

$$\text{Demand of skills in a skill cluster} = \frac{\text{Total count of skills for a given cluster in a year}}{\text{Total count of skills in a year}} * 100$$

In this chapter, data from five years (2019-2023) was analysed to provide a time-series view of the skills requirement for job roles. Additional analysis focused on the top 10 in-demand skills within each job role's three most in-demand skill clusters were also included.

E. Performing data analysis for Apps & Tools via work functions

In a given year, to derive the **demand of Apps & Tools used by a work function**, the total count of Apps & Tools tagged to a work function was divided by the total count of Apps & Tools and presented as a percentage.

The same calculation for demand was done for AI-related Apps & Tools.

$$\text{Demand of Apps \& Tools in a work function} = \frac{\text{Total count of Apps \& Tools for a given work function in a year}}{\text{Total count of Apps \& Tools in a year}} * 100$$

Analysis was also done on the top 10 demanded Apps & Tools for any particular job role. This refers to the top 10 mentioned Apps and Tools based on the number of their mentions in job postings for that particular job role.

F. Measurement of job demand and change of job demand

Job demand was derived by dividing the number of job posting of a given job role in year, by the total number of job posting in the same year and presented as a percentage.

$$\text{Job demand} = \frac{\text{Number of job posting of a given job in a year}}{\text{Total number of job postings in a year}} * 100$$

Change of job demand was derived by using **Compound Annual Growth Rate (CAGR)** based on job postings from 2020 to 2023. This resulted in a percentage that reflected the growth in employers' demand for the job compounded over four calendar years.

$$\text{Job demand}_{\text{CAGR}} = \left(\frac{\text{Number of job posting of a given job in 2023}}{\text{Number of job posting of a given job in 2020}} \right)^{\left(\frac{1}{3}\right)} - 1$$

Analysis methodologies applied in Chapter 2: Priority Skills Forecast and Movement

A. Definition of Priority Skills

Priority skills are a basket of skills that are deemed as currently in-demand, seeing positive growth or demanded by more than one unique job role. These skills were derived from SSG's National Jobs-Skills Intelligence engine and validated with expert input from industry, academia and sector agencies. In the various charts within this chapter, SSG chose to spotlight skills whose demand grew in recent years, rather than just skills with a large, existing demand. The spotlighted skills were more likely to see shortages in the present and near future.

B. Measurement of Demand and Transferability of a Skill

The **demand of a skill** represents the proportion of a given priority skill within the total number of skills listed in the economy group. The formula was derived as:

$$\text{Demand of skill} = \frac{\text{Count of job posting for a given priority skill in year}}{\text{Count of job posting for all priority skills in year}} * 100$$

Transferability of a skill refers to the number of unique job roles that require the skill in that year. In this segment, transferability was calculated year by year based on job postings from 2022-2024 and for 2025 using the Skills Forecasting methodology. The formula was derived as:

$$\text{Transferability of Skill} = \sum_{i=1}^n \text{Unique SSOC that required this skill}_{(\text{year})}$$

C. Defining Skills Forecasting

Skills Forecasting refers to the predicted demand and transferability for 2025 generated by a machine-learning model.

The machine-learning model for Skills Forecasting was trained with historical demand data or transferability and employed Linear Regression to predict the value of all skills for the year 2025. The model used a sliding window approach, incorporating the average demand or transferability of each skill over a three-month window, extending up to two years on a monthly basis, thus creating a total of 24 windows as predictors. Additionally, the model took in the embedding of each skill title using Word2Vec, which were also used as part of the set of predictors.

D. Understanding the Skills Matrix

Priority skills were plotted on a scatter plot with their demand as the vertical axis and transferability (i.e. number of unique job roles requiring that skill) as the horizontal axis for each given year. Then, the priority skills were categorised using the 70th percentile mark as a benchmark for both axes, resulting in four quadrants of skills: In-demand quadrant, Up-and-coming quadrant, Niche quadrant, and Transferable quadrant. A given priority skill can be plotted on the Skills Matrix for a given year, into one of the four quadrants. Repeating this for a number of consecutive years, the position of the skill on the Skills Matrix may change, resulting in the skill movement being observed across years.

Analysis methodologies applied in Chapter 3: Creative Industries Skills Spotlight

A. Scoping of Creative professionals

Creative professionals include job roles whose primary responsibilities involve creative work such as art, design, media, marketing, and event management. They may be working in companies whose core business is in creative fields or creative departments within businesses in other industries.

A total of 71 creative job roles, at SSOC level 5 were identified, of which job posting data of 45 creative job roles were available for analysis.

B. Performing data analysis of Skills

Based on the job posting data of creative job roles, the demand of a skill cluster was derived by dividing the total count of skills within each cluster from job postings each year, by the total count of skills across all six clusters based on job postings for the same year.

Data from five years (2019-2023) was analysed to provide a time-series view of the job market, comparing skills and economic trends.

C. Performing data analysis for Apps & Tools via work functions

The computation of the Apps & Tools was based on the methodology used in Chapter 1.

For analysis on Apps & Tools used by Creative professionals, only job posting data of Creative professionals were used.

For analysis on Apps & Tools used by non-Creative professionals, job posting data of non-Creative SSOCs were used, and those performing work relating to art, design, marketing, and media and communications were further examined.

D. Performing data analysis on transferability of creative skills

Creative skills were identified based on skills specific to sectors in Design, Media, Built Environment (Architect) and Tourism (content and experience creation) as well as skills that were tagged to creative job roles relating to Marketing, Branding, Public Relations, Attraction and Event Management.

The transferability of creative skills was based on the methodology used in Chapter 2.

E. Poll on Creative professionals' perception of GenAI to their work

The poll was conducted to capture the general sentiments of Creative professionals on the use of GenAI at work, perceived benefits, potential applications and considerations integrating into work.

It was administered online from 24 Jul to 17 Sep 2024 by inviting Creative professionals through various channels including LinkedIn and professional communities. A total of 104 responses were collected, of which 87 indicated that they were professionals from Arts, Marketing and Media, and their responses were tabulated accordingly.

Analysis methodologies applied in Chapter 4: Explore Pathways to In-Demand Jobs (In partnership with Workforce Singapore)

A. Identification of roles suitable for career transition

Job roles that have a) good wages, b) demanded by employers currently, and c) have good growth were identified as jobs suitable for individuals to transition into:

- Good wages are defined as S\$3,300 and above. S\$3,300 was selected as a benchmark as it is 65% of Singapore's Median wage of S\$5,197. The median gross monthly income is obtained from Comprehensive Labour Force Survey, Manpower Research and Statistics Department, MOM. The median gross wage for each job role for this study is from the Occupational Wage Survey conducted by Manpower Research & Statistics Department, Ministry of Manpower.
- Job demand was derived by dividing the number of job posting of a given job role in year, by the total number of job posting in the same year and presented as a percentage.

$$\text{Job demand} = \frac{\text{Number of job posting of a given job in a year}}{\text{Total number of job postings in a year}} * 100$$

Good demand is set at 25th percentile of market share, to bring focus on job roles that employers are actively seeking to fill.

- Change of Job Demand (CAGR) is calculated using the Compound Annual Growth Rate (CAGR) of job postings for each job role.

$$\text{Job demand}_{\text{CAGR}} = \frac{\text{Number of job posting of a given job in 2023}^{\frac{1}{3}}}{\text{Number of job posting of a given job in 2020}} - 1$$

Using the CAGR of all job postings from 2020 to 2023, we considered job roles with CAGR of 25th percentile and above as those that are generally maintaining their importance in the job market, and not experiencing a big drop in demand by employers.

Upon applying these filters, 146 job roles were identified as suitable for career transition. While there are 1002 job roles with unique SSOC codes, only 342 unique SSOCs have wage and job demand data relevant for this analysis. Only 5-digit SSOCs with wage information were used for this study.

Explanatory Notes

The Occupational Wage Survey covers a representative sample of private sector establishments each with at least 25 employees. The occupational wage data presented are for full-time resident employees only. This provides a more meaningful basis for comparison of wages across occupations.

Monthly gross wage refers to the sum of the basic wage, overtime payments, commissions, allowances, and other regular cash payments. It is before the deduction of employee Central Provident Fund (CPF) contributions and personal income tax and excludes employer CPF contributions, bonuses, stock options, other lump sum payments and payments-in-kind.

Analysis methodologies applied in Chapter 4: Explore Pathways to In-Demand Jobs (In partnership with Workforce Singapore)

B. Identification of potential career pathways for roles suitable for career transition

Career pathways comprise of the original role (i.e. the user's current job role), followed by a first destination role, and a second destination role. The first destination role is a role identified to be potentially suitable role for career transition as mentioned in (A) on page 75.

These pathways were identified using SSG developed proprietary knowledge graph¹ connecting data on jobs, skills and training courses.

Job transitions were identified based on the following criteria:

- a) Job roles with many skills in common: This was measured in two ways, and in both cases greater importance was placed on skills that were considered more “unique”² to a job role:
 - i. The ratio of skills common to both job roles, over all skills required by both job roles, taking into account the importance of a skill to an occupation (weighted

Jaccard similarity). This ratio determined the ranking of the adjacent job roles that an original role can transition into. The larger the number, the higher the similarity between the two jobs, which indicated that the original role has more of the skills needed for the next job role.

- ii. The absolute number of skills needed for both job roles

- b) Count of Job postings of the destination job. To be included, jobs must have more than 10 postings in 2023 data used for this report.
- c) **Historical transitions** between jobs: This was based on whether there were actual transitions between two SSOCs in MOM's Occupational Employment Dataset (OED)³ from Dec 2020 to Sep 2022.

1 Knowledge graphs represent networks of real-world entities—in this case, jobs, skills and courses—and the relationships between them.

2 This “uniqueness” was calculated using the Term Frequency – Inverse Document Frequency (TFIDF) method that is commonly used in Natural Language Processing. A skill was determined to be more “unique” to a particular occupation if it was commonly that occupation, but not for other occupations in the same SSOC 3-digit cluster.

3 Occupational Employment Dataset (OED) is an ongoing registry where organisations in Singapore are required to provide the job and work location details of all employees.

Glossary

Term	Definition
Applications and Tools (Apps & Tools)	Digital and technology solutions programmes, in software or app formats, used in the application of skills to perform job tasks .
Artificial Intelligence (AI)	A branch of computer science focused on creating systems that can perform tasks typically requiring human intelligence. These tasks include learning, problem-solving, pattern recognition, decision-making, and language processing. AI systems use algorithms and data to simulate cognitive functions, enabling machines to adapt to new inputs and perform human-like tasks.
AI-related Apps & Tools	Apps & Tools that contribute to the creation, development, or deployment of AI, leverages AI capabilities, or associated with AI in general .
Care Economy	An economy that is based on a professional cluster of jobs and skills that provides care and support services involved in the nurturing and teaching of current and future populations .
Change of job demand	Change in number of job postings demanded over four years, from 2020 to 2023 .
Compound annual growth rate (CAGR)	The average annual growth rate over a specified period of time longer than one year. It represents one of the more accurate ways to calculate anything that can rise or fall in value over time.
Critical Core Skills	A unique set of 16 core skills identified by Singapore employers as the most critical to thrive in the future economy .
Demand of Apps & Tools in a work function	The demand of Apps & Tools used by a work function over the total demand of all Apps & Tools extracted from job postings in a year.

Glossary

Term	Definition
Demand of a skill	The yearly demand of a given priority skill refers to that skill's demand, relative to the aggregate demand of all the skills for its respective economy, based on job postings data in a year.
Demand of a skill cluster	The six skill clusters are: Business and Financial Management, Critical Core Skills, Engineering and Manufacturing, Information Technology and Data Management, Operational Excellence, and Organisational and People Management. The demand of a skill cluster is the demand of skills within a given cluster relative to the total count of skills in all six skill clusters based on job postings in a year.
Digital economy	An economy that is based on digital computing technologies, based on interconnecting people, organisations and machines through the Internet, mobile technology and the Internet-of-Things (IoT).
Green Economy	An economy that strives to achieve environmental, economic and social outcomes to take care of the environment and use limited resources as efficiently and sustainably as possible .
Growth of a job role	Defined as the increase in absolute number of job postings for a particular job role over time.
Historical Transition	These are actual transitions from one job role to another.

Glossary

Term	Definition
Job demand	The demand of a job role expressed in terms of the number of job postings of a given job role in year, relative to the total number of job postings in the same year.
Priority skills	Priority skills are a basket of skills that are deemed as currently in-demand, seeing positive growth or demanded by more than one unique job role. These skills have been validated by industry expert, academia and sector agencies.
Skill cluster	A group of skills that is associated with broad business functions and the critical core skills. These skill clusters reflect the skill requirements needed across job roles in the Singapore economy.
Skills similarity	Ratio of skills in common to all skills required by two comparing occupations.
Transferability of a skill	Transferability of a given priority skill refers to the number of unique job roles that require the skill in a given year from job posting data.
Unique job roles	Unique job roles are job roles at Singapore Standard Occupation Classification (SSOC) five-digit that have distinct tasks, skills and functions.

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Skills Mentioned

Skill Title	Skill Description
<i>Account Management</i>	Manage, maintain and grow the sales and relationships with a specific customer or set of accounts. This includes in-depth customer engagement, relationship-building and provision of quality solutions and service to address customers' needs efficiently and generate revenue.
<i>Advanced Generative AI Technology</i>	Exploring the advanced techniques, cutting-edge developments, and ethical considerations in the field of Generative AI technology.
<i>Application Development of Generative AI Models</i>	Designing and building applications around generative AI models, integrating the language processing capabilities towards specific use cases and domains.
<i>Apply Teamwork In The Workplace</i>	Work as a team and participate in team activities in their work area
<i>Apply Workplace Safety and Health in Construction Sites</i>	On completion of this unit, learners will have the skills and knowledge in applying workplace safety and health practices to ensure the safety of oneself and others at work in the construction Industry by identifying own roles and responsibilities in construction job, following safe work procedures, adopting relevant risk control measures when working at heights and responding to fire emergencies
<i>Architecture Design</i>	Utilise holistic design approaches for the conceptualisation, development and enhancement of design solutions
<i>Assessment Design and Implementation</i>	Design and implement assessment methods and tools to evaluate learner progress, taking into account the learning objectives of the specific art form and incorporating showcase style assessment opportunities where appropriate and as relevant to the art form, such as performances, portfolios, and exhibitions.
<i>Brand Management</i>	Co-create the organisation's projected brand and reputation with the customer, consider customer's perspectives and the organisation's desired image and priorities. This also includes the development and execution of branding campaigns, public relations and reputation management strategies to sustain or enhance the desired brand

Skills Mentioned

Skill Title	Skill Description
<i>Business Networking</i>	Establish mutually beneficial relationships with business stakeholders, potential clients and customers
<i>Business Opportunities Development</i>	Develop and implement plans to enhance organisation's business performance and growth
<i>Business Presentation Delivery</i>	Perform required tasks to prepare and present information in various business settings involving preparation, understanding of audience, delivery and tailoring of messages to be conveyed
<i>Business-to-Consumer Management</i>	Execute customer management strategies to complete sales of products and services to the satisfaction of customers
<i>Carbon Footprint Management</i>	Quantify and reduce the organisational carbon footprint
<i>Care Assistance</i>	Provide care, assistance and training in Activities of Daily Living ADLs
<i>Case Reporting and Documentation</i>	Integrate, organise and document the essential facts and observations of a case in a manner that is clear, coherent and in the best interest of the client
<i>Case and Care Planning</i>	Use different models and approaches for developing case and care plans to support clients' needs in accordance to organisational practices
<i>Casework Evaluation</i>	Evaluate effectiveness and efficiency of casework interventions
<i>Casework Intervention</i>	Provide systematic approaches to casework interventions

Skills Mentioned

Skill Title	Skill Description
<i>Child Safety and Protection</i>	Implement safety standards and procedures, and appropriate courses of action to ensure the safety and protection of children
<i>Civil and Structural Engineering Management</i>	Manage the design, technical specification, selection, modification and troubleshooting of civil structures and systems to provide civil and structural engineering discipline support to construction, maintenance and project teams
<i>Client Investment Suitability</i>	Determine client suitability, financing needs, risk management requirements, investment objectives and propose relevant sales opportunities
<i>Clinical Supervision</i>	Support the professional development of counsellors in advancing therapeutic, casework and counselling skills, and maintaining high standards of professional practice through a myriad of supervisory modalities and models by providing a supportive learning environment for supervisees
<i>Clinical Support Duties and Procedures</i>	Provide assistance with clinical duties and simple clinical procedures
<i>Clinical Support for Patient Service Associates</i>	Assist with clinical duties or perform simple clinical procedures during consultation as instructed by registered practitioners and according to organisational guidelines and procedures
<i>Community Partnership</i>	Establish and foster partnerships with community stakeholders through a culture of collaboration to develop mutually beneficial programmes for children and families
<i>Conceptual Thinking</i>	Analyse and synthesise information by identifying key issues, perceiving unseen patterns and trends and deducing connections between issues to develop relevant ideas and solutions

Skills Mentioned

Skill Title	Skill Description
<i>Conflict Management</i>	Build consensus, maintain the best interests of the organisation and utilise knowledge of conflict management techniques to diffuse tensions and achieve resolutions effectively
<i>Construction Technology</i>	Identify and apply the most suitable and effective construction processes and technologies to achieve project objectives
<i>Continuing Professional Development Management</i>	Facilitate the implementation of continuing professional development plans within the organisation to extend, update and maintain the technical competences of professionals
<i>Continuous Improvement Management</i>	Apply continuous improvement strategies to improve products, services or processes across the organisation
<i>Contract Administration and Management</i>	Develop contracts to determine suitable conditions of contract, and optimise the contract administration and management process taking into consideration the nature of the project
<i>Contract Preparation, Evaluation, Negotiation and Tendering</i>	Manage contract creation, evaluation , negotiation, tendering to maximise operation and financial performance of an organisation
<i>Contracts and Agreement</i>	Analyse and manage intellectual property contracts and agreements
<i>Core Principles of Generative AI</i>	Comprehending the concepts, applications, and implications of Generative AI models.
<i>Corrective Maintenance Management</i>	Conduct diagnostic tests and inspections on assigned plants, equipment and systems, and develop plans for corrective maintenance, repairs and overhauls

Skills Mentioned

Skill Title	Skill Description
<i>Counselling Intervention</i>	Conduct counselling interventions to address socioemotional, mental, and social challenges
<i>Crisis Situations Management</i>	Identify crisis response and recovery activities as well as implement the recovery and business continuity strategies to minimise the impact of disruptive events to the organisation
<i>Customer Acquisition Management</i>	Develop customer acquisition strategies as well as foster customer relationships to attract new customers
<i>Customer Experience Management</i>	Compile and analyse information gathered through various channels and manage communication across customer touch points to ensure a consistent and pleasant customer experience
<i>Customer Management</i>	Manage customers with the goal of improving business relationships with customers and achieving service requirements
<i>Cyber and Data Breach Incident Management</i>	Detect and report incidents, identify affected systems and user groups, trigger alerts and announcements to relevant stakeholders, and ensure efficient resolution of situations
<i>Data Governance</i>	Develop, implement and practice guidelines, laws, and regulations across the organisation for the handling of data at various stages in its lifecycle as well as the provision of advice on proper data handling and resolution of data breaches
<i>Design Creation and Development</i>	Utilise relevant design approaches for the conceptualisation, development and enhancement of design solutions
<i>Digital and Physical Prototyping</i>	Construct design concepts, either digitally or physically, to develop deeper understanding of the designs and test their usability and functionality

Skills Mentioned

Skill Title	Skill Description
<i>Document Control</i>	Implement documentation policies to facilitate the referencing of information for processes, systems and equipment, and to comply with regulatory requirements
<i>Effective Client Communication</i>	Demonstrate effective communicative skills when communicating with clients and caregivers
<i>Electrical, Electronic and Control Engineering</i>	Manage and monitor electrical, electronic and control engineering systems
<i>Engineering Drawing and Design Specifications</i>	Create design specifications and technical drawings to guide installation and construction works
<i>Engineering, Procurement and Construction Management</i>	Manage engineering design, procurement and construction for new process plants and/or plant expansion projects
<i>Ethical and Governance Considerations in Generative AI</i>	Grasping foundational ethical concerns and progressing to integrating ethics into AI projects, leading to advanced insight, and framework creation for responsible AI development.
<i>Ethical Conduct and Professional Integrity</i>	Understand the professional conduct, ethics and values and comply with the relevant legislation to uphold the integrity and reputation of the profession
<i>Events Planning and Management</i>	Develop, manage and execute event plans according to agreed timelines and budgetary requirements
<i>Food and Beverage Services</i>	Prepare for service of food and beverages to customers in accordance with the service delivery standards of the organisation

Skills Mentioned

Skill Title	Skill Description
<i>Generative AI Evaluation</i>	Assess the quality, performance, and potential impact of Generative AI models across various domains and applications.
<i>Generative AI Model Development</i>	Apply deep learning expertise and statistical inference knowledge to design, implement, train, and evaluate Generative AI models for various tasks and applications, assessing the quality, performance, and potential impact of Generative AI models across various domains and applications.
<i>Geotechnical Engineering Management</i>	Manage the design, technical specification, selection, modification and troubleshooting of geotechnical equipment, structures and systems to provide geotechnical engineering discipline support to construction, maintenance and project teams
<i>Green Facilities Management</i>	Manage facility operations and maintenance to minimise environmental impact and operational costs efficiently
<i>Greenhouse Gas Accounting</i>	Assess, quantify, and verify greenhouse gas emissions produced by an organisation or organisational activities, to support the monitoring and management of an organisations carbon footprint. Ensure compliance with environmental standards and regulations, and develop strategies for the reduction of emissions and mitigation of climate change impacts.
<i>Group Work Assessment and Planning</i>	Perform client assessments, develop and plan group work processes and specialised intervention strategies to achieve effective outcomes
<i>Hospitality Venue Inspection</i>	Organise and conduct property site inspections to introduce facilities and features of property to clients for the purpose of closing sales
<i>Idea Generation And Selection</i>	Facilitate discussions, seek opinions from others, brainstorm ideas to realise business opportunities and improve current working conditions

Skills Mentioned

Skill Title	Skill Description
<i>Inspection Engineering Management</i>	Manage fixed equipment and piping inspection schemes, materials selection, construction, corrosion control, condition and fitness-for-service through on-stream, risk-based monitoring programmes and downtime inspections to provide inspection engineering technical support to maintenance, engineering design and project teams
<i>Integrated Marketing</i>	Develop and execute marketing plan on and across various channels and platforms, as well as track customers' responses to and effectiveness of marketing communications on these channels. This also includes the integration of traditional and digital marketing channels and techniques where applicable
<i>Intervention Implementation</i>	Perform the appropriate interventions based on client's needs, conditions and care plan
<i>Learning and Development</i>	Develop knowledge, skills and abilities of self or employees to support the capability development needs of the organisation
<i>Legal and Compliance Management</i>	Manage the adherence to applicable legislations and compliance standards
<i>Maintenance Strategy Development</i>	Develop a corrective and preventive maintenance strategy
<i>Manage Workplace Safety and Health in Construction Sites</i>	Manage the worksite safety and health management system and ensuring safety and health measures are put in place in construction activities.
<i>Market Research</i>	Formulate market research frameworks, as well as develop market research study objectives, market research plans and methodologies to analyse market trends and developments to forecast emerging market needs

Skills Mentioned

Skill Title	Skill Description
<i>Market Trend Analysis</i>	Plan and conduct marketing and digital research and analysis to uncover market, customer and competitor trends in order to extract useful business insights. This also includes the evaluation of marketing activity effectiveness and development of ways to optimise marketing efforts
<i>Marketing Campaign Management</i>	Develop evaluation strategies for marketing campaign effectiveness and analyse data to provide recommendations for improvements in future marketing campaigns
<i>Marketing Management</i>	Manage organisation's marketing plans
<i>Marketing Operations Management</i>	Execute marketing tasks and activity operations
<i>Marketing Strategy Development and Implementation</i>	Develop sales and marketing strategies and operational plans and evaluate performance outcomes to identify improvements to enhance sales performance and achieve organisational strategic objectives
<i>Meetings, Incentives, Conferences and Exhibitions (MICE) Content and Experience Development and Delivery</i>	Manage the content and experience for Meetings, Incentives, Conferences and Exhibitions MICE events, including programme and itinerary development, identification of key speakers and partners, and liaising with exhibitors regarding content curation, exhibition specifications and exhibition layouts
<i>Monitoring and Surveillance</i>	Develop and conduct checks and observations to assess adherence to statutory regulations and standards, thoroughness of compliance controls and monitor for irregular activities.
<i>Order Fulfilment Administration</i>	Administer receiving, processing, delivery and optimisation processes for orders in order to support business and customer requirements

Skills Mentioned

Skill Title	Skill Description
<i>Organisational Design</i>	Develop and facilitate the implementation of organisational design to ensure its effectiveness and alignment with stakeholders' priorities
<i>Organisational Policy Development for Workplace Safety and Health</i>	Develop organisational Workplace Safety and Health policies to ensure compliance with prevalent national regulations and practices
<i>Perform Manhole Safety Assessment</i>	Apply safe work practices and regulatory requirement on manhole and/or confined space, conduct risk assessment to identify and assess hazards, perform gas testing and record and interpret the gas measurement results activities in construction and related industries
<i>Print Management</i>	Manage print production, monitor progress and apply quality control procedures to ensure compliance with project specifications as well as contractual agreements
<i>Product Demonstration</i>	Develop knowledge of the range and price of the organisation's products and services, as well as present and demonstrate the use and application of products and services to customers
<i>Product Marketing and Branding</i>	Develop and implement product marketing strategies by leading sales data collection, segmentation, market research, product marketing idea development and optimisation. Implement product marketing strategies and recommend changes based on desired product and/or corporate positioning and branding
<i>Programme Evaluation</i>	Evaluate the effectiveness and efficiency of programmes, and contribute to continuous programme improvement
<i>Programme Planning</i>	Develop programming strategies and plans to manage the placement of the organisations content on linear and non-linear platforms and to maximise the reach and monetisation potential of the content

Skills Mentioned

Skill Title	Skill Description
<i>Programme Quality and Audit Management</i>	Ensure the quality of services adheres to standards and develop initiatives to enhance quality through systematic review of programmes and services against criteria and the implementation of changes
<i>Programme and Project Management</i>	Manage the implementation and development of programmes and projects to facilitate achievement of organisations objectives and growth.
<i>Project Coordination</i>	Coordinate project activities and workflows in collaboration with project teams and relevant stakeholders, as determined by project plans, to fulfil expected project outcomes and objectives
<i>Project Execution and Control</i>	Implement projects in accordance with project plan and deliverables, and monitoring and controlling processes performed to influence project outcomes
<i>Project Management</i>	Execute projects by managing stakeholder engagement, resources, budgets and resolving problems
<i>Project Plan</i>	Develop project plans and manage project risks using appropriate project management tools
<i>Project Timeline</i>	Determine project requirements and develop timelines to ensure timely project delivery
<i>Prompt Engineering</i>	Crafting effective prompts to shape outputs in AI models, evolving from basic construction to nuanced and strategic input design to create prompts that optimize AI model performance, combining different methodologies to generate innovative and effective prompts
<i>Quality Control Management</i>	Establish and implement quality control systems and procedures to ensure the quality of products meet desired levels of standards and compliance at all stages

Skills Mentioned

Skill Title	Skill Description
<i>Quality System Management</i>	Coordinate and direct the organisations activities to meet customer and regulatory requirements as well as identify opportunities for improvement.
<i>Regulatory Submission and Clearance</i>	Manage the processes for obtaining the required project approvals, permits, and the compliance to regulatory conditions
<i>Report Writing</i>	Present specific information and evidence in a clear and structured format
<i>Research</i>	Research on a concept or idea to provide inputs for content development
<i>Retail Administration</i>	Manage and validate petty cash transactions, as well as document hours worked for each employee
<i>Risk Assessment</i>	Perform assessment of risks, including fraud risks, through understanding the clients business
<i>Sales And Business Development</i>	Manage business development efforts to generate new business or upsell and increase business with existing accounts
<i>Sales Closure</i>	Perform numerical calculations and execute selling strategies to complete sales of products and services to the satisfaction of customers
<i>Sales Delivery</i>	Manage sales delivery activities to achieve sales for the organisation through new and existing clientele
<i>Sales Management</i>	Develop, monitor, and implement sales plan to achieve business objectives
<i>Sales Strategy</i>	Define sales strategies and market penetration plans for the retail target segment

Skills Mentioned

Skill Title	Skill Description
<i>Sales Target Management</i>	Evaluate and monitor sales target and performance to plan and initiate actions to achieve excellence in sales delivery
<i>Sales Trends and Opportunities Analysis</i>	Analyse sales trends and opportunities to manage inventory, evaluate sales strategies and pursue new sales leads to meet organisational sales and marketing objectives
<i>Service Partnerships</i>	Collaborate with partners to deliver and enhance the customer experience
<i>Service Planning and Implementation</i>	Develop and implement strategies and plans for the service operations
<i>Service Quality Management</i>	Manage, develop and review the quality with which services are provided to clients
<i>Social Service Programme Development</i>	Develop social service programmes using intervention concepts and frameworks that serve the needs of the clients and community
<i>Social Service Programme Implementation</i>	Implement effective social service programmes that serve the needs of the clients and community
<i>Stakeholder Management</i>	Manage stakeholder expectations to ensure continuous levels of engagement by identifying and addressing needs, setting service standards and resolving issues in accordance with organisational procedures
<i>Strategy Development and Implementation Management</i>	Develop and implement organisational strategic plans and provide direction to the organisation
<i>Strategy Execution</i>	Develop and implement plans to achieve organisational and departmental strategies and goals

Skills Mentioned

Skill Title	Skill Description
<i>Strategy Planning</i>	Develop organisational strategies and policies by analysing the impact of internal and external influencing factors and seeking consultation from relevant stakeholders.
<i>Supervise Metal Scaffold Erection</i>	Supervise scaffold erection and dismantling work activities in construction and other industries.
<i>Supervise Workplace Safety and Health for Formwork Construction</i>	Supervise formwork activities which include the construction, erection, alteration or dismantling and inspection of formwork structures at any worksite
<i>Support Service to Children and Youths</i>	Provide support services to children and youths to ensure their overall well-being
<i>Support Service to Seniors</i>	Provide support services to seniors, their families and caregivers by identifying and understanding their needs and concerns
<i>Sustainability Assurance</i>	Evaluate and verify the accuracy, relevance, and completeness of an organisations sustainability reporting, ensuring alignment with established regulatory standards, environmental goals, social responsibility, and economic outcomes. Enhance the transparency, credibility, and trust in an organisations sustainability reporting, ensuring that informed decisions can be made.
<i>Sustainability Management</i>	Plan, develop and roll out of an organisation-wide sustainability strategy. This includes the assessment of the organisation's utilisation and/or consumption of energy and other resources, vis-a-vis the availability and stability of supply sources and external best practices and standards in sustainability. This also includes the on-going monitoring and tracking of energy and/or resource-consumption over time, to identify impact on the organisation's internal and external environment as well as potential improvements in energy- or resource-efficiency.
<i>Sustainable Engineering</i>	Design, construct and operate engineering systems and assets to optimise energy management and enhance environmental performance

Skills Mentioned

Skill Title	Skill Description
<i>Team Management</i>	Manage staff performance by developing systems and processes to ensure key performance indicators are met
<i>Technical Drawing</i>	Manage the composition of precise and detailed drawings and specifications that visually communicate how designs function, or are produced, to guide the construction
<i>Technical Inspection</i>	Execute formal inspection exercises to ensure quality, safety, and reliability adhering with technical specifications and compliance requirements
<i>Trade Processing, Fund Settlement and Treasury Transaction Processing</i>	Demonstrate the ability to validate appropriate fund terms and transfers, monitor and review the smooth delivery of securities to buyers, perform settlement, rate verification, reconciliation and post-trade confirmation activities including reporting transactions that have not been cleared or are unusual
<i>Use of Generative AI Models</i>	Utilizing Generative AI models, starting from basic tasks to handling diverse and complex tasks proficiently, showcasing adaptability, optimization, and strategic model selection.
<i>Utilities Management</i>	Develop plans to meet manufacturing utilities and energy requirements while conserving and managing the use of energy and utilities by the facility
<i>Volunteer Programme Management</i>	Manage volunteer programmes, operations and logistics and develop volunteer programmes for the organisation
<i>Workflow Management</i>	Design, construct and operate engineering systems and assets to optimise energy management and enhance environmental performance

Skills Mentioned

Skill Title	Skill Description
Workplace Optimisation	Plan and manage the work spaces to encourage collaboration and build organisation culture
Workplace Safety and Health Supervision in Tunnelling Construction	Apply appropriate control and safety measures when supervising deep excavation and tunnelling construction works common safety
Youth Outreach	Interact and engage with youths in diverse settings through platforms and activities that interest them

Apps & Tools Mentioned

Apps & Tools Title	Apps & Tools Description
<i>AWS Cloud9</i>	Cloud9 IDE is an Online IDE, published as open source from version 2.0, until version 3.0. It supports multiple programming languages, including C, C, PHP, Ruby, Perl, Python, JavaScript with Node.js, and Go.
<i>AdminStudio</i>	AdminStudio, which is a single software solution for application packaging, application virtualization, Windows migrations and reliable software deployment.
<i>Adobe Illustrator</i>	Adobe Illustrator is a vector graphics editing software that allows users to create and edit graphics, illustrations, and designs. It is commonly used by graphic designers, artists, and illustrators to create marketing materials, logos, branding assets, and more. The software offers various tools and features to manipulate shapes, colors, and typography to achieve the desired outcome. It also enables users to export their designs in various file formats compatible with other design software or printing presses.
<i>Adobe Photoshop</i>	Adobe Photoshop is a powerful software used for editing and manipulating digital images. It is an essential tool for graphic designers, photographers, and other professionals who work with visual media. With its wide range of features and tools, Adobe Photoshop allows users to enhance, retouch, or transform images to match their creative vision. Some of its core features include layers, filters, selection tools, and adjustment layers. It is a key skill to have in many creative and technical fields.
<i>As-Built Report</i>	As Built Report is an open source configuration document framework which utilises Microsoft PowerShell to produce as-built documentation in multiple document formats for multiple vendors and technologies. The framework allows users to easily generate clear and consistent documentation, for any environment which supports Microsoft PowerShell and/or a RESTful API.
<i>AutoCAD</i>	AutoCAD is a specialized skill that refers to a computer-aided design CAD software used by architects, engineers, and designers to create 2D and 3D drawings and models of buildings, products, and mechanical parts. This software allows users to manipulate geometry and create precise measurements and visualizations of their designs. AutoCAD is widely used in industries such as construction, manufacturing, and architecture.

Apps & Tools Mentioned

Apps & Tools Title	Apps & Tools Description
Automation Studio	Automation Studio is a software used for designing, simulating, and programming automation systems. It provides engineers and technicians with a tool to create hydraulic, pneumatic, electrical, and automation systems. The software allows for simulation and visualization of systems before they are built, saving time and resources while reducing errors. With Automation Studio, users can design and troubleshoot complex systems, create customized libraries, and automate repetitive tasks.
BPCS	Business Planning and Control System (BPCS) is an Enterprise Resource Planning (ERP) software designed for an OS400 system.
C	C is a general-purpose computer programming language. It was created in the 1970s by Dennis Ritchie, and remains very widely used and influential. By design, C's features clearly reflect the capabilities of the targeted CPUs. A
C++	C++ is a general-purpose programming language created by Danish computer scientist Bjarne Stroustrup as an extension of the C++ programming language, or "C++ with Classes". The language has expanded significantly over time, and modern C++ now has object-oriented, generic, and functional features in addition to facilities for low-level memory manipulation.
Campaign Monitor	Campaign Monitor is a cloud-based email marketing software that allows businesses and individuals to create, send, and track customized email campaigns. It helps organizations build and maintain strong relationships with their subscribers and customers through targeted and personalized email communications. The software offers a range of features such as email list management, drag-and-drop email builder, email automation, and detailed performance analytics.
CaseMaps	CaseMaps is a specialized software designed for legal professionals that helps in organizing case-related information and making it easier to analyze and present in court. It allows for the creation of visual timelines, maps, and charts to help lawyers and paralegals better understand and present complex information. CaseMaps also provides collaboration tools to help legal teams work together more efficiently, and it integrates with other legal software to streamline workflows.

Apps & Tools Mentioned

Apps & Tools Title	Apps & Tools Description
<i>DoubleClick Digital Marketing</i>	DoubleClick Digital Marketing is a software platform that supports digital advertising campaigns. It offers solutions for managing multiple campaigns, creating and targeting ads, measuring and analyzing performance, and optimizing ad placements across various channels and devices. The platform includes tools for display advertising, search advertising, video advertising, and programmatic advertising, and it enables advertisers to reach audiences across the web and mobile apps.
<i>Facebook Insights</i>	Facebook Insights is a tool used by businesses and organizations to analyze the performance of their Facebook pages. It provides data and metrics on post engagement, page likes, reach, demographics, and more. Facebook Insights allows businesses to better understand their audience and how their content is performing, so they can optimize their social media strategy to improve engagement and ROI.
<i>Generalized Audit Software</i>	Generalized Audit Software GAS is a type of software that enables auditors to automate and perform various auditing procedures on large volumes of data quickly and accurately. It is designed to extract, manipulate, and analyze data from various sources such as spreadsheets, databases, and accounting software. GAS can perform tasks such as data validation, statistical analysis, fraud detection, and risk assessment. It is a specialized skill that requires auditors to have a strong understanding of data analytics and computer programming. With the help of GAS, auditors can improve audit efficiency and effectiveness, minimize errors, and reduce manual efforts.
<i>Google Ads</i>	Google Ads is an online advertising platform created by Google, which allows businesses to create and run ads on Google's network. It involves keyword research, ad creation, bidding and budget management, and ongoing analysis and optimization. Specialized skills related to Google Ads include data analysis, strategic planning, and campaign management.
<i>Google Analytics</i>	Google Analytics is a web analytics service that allows website owners or marketers to track and analyze their website's visitors and user behavior. This tool provides valuable insights into the performance of a website and enables site owners to make data-driven decisions to improve their site's user experience, content, and marketing efforts. A person with a Specialized Skill in Google Analytics would have expertise in using this software to collect, analyze, and interpret website data to its full potential.

Apps & Tools Mentioned

Apps & Tools Title	Apps & Tools Description
IBM Case Manager	IBM Case Manager is a software platform that offers organizations a powerful set of tools for designing, deploying, and managing business processes. This specialized skill involves expertise in configuring and customizing the platform to meet specific business needs, and requires knowledge of programming languages such as Java and JavaScript, as well as experience with database management and server administration. IBM Case Manager can be used to streamline document management, automate workflows, and improve collaboration and communication within an organization.
Informatica	Informatica is a specialized software skill that involves the use of a data integration platform for extracting, transforming, and loading data into various systems. It is used to manage and manipulate large amounts of data across different sources and formats, making it easier for organizations to analyze and use their data. Informatica skills are essential for data analysts, data engineers, and other professionals who work with large datasets.
Instagram	Instagram is a social media platform that allows users to share photos and videos with their followers. Being able to use Instagram effectively requires a specialized skill set, including knowledge of how to create engaging and visually appealing content, use hashtags and other tools to increase visibility, and build a following. Other related skills might include social media marketing, content creation, and graphic design.
Java	Java is a high-level, class-based, object-oriented programming language that is designed to have as few implementation dependencies as possible.
JavaScript	JavaScript , often abbreviated JS, is a programming language that is one of the core technologies of the World Wide Web, alongside HTML and CSS. As of 2022, 98 of websites use JavaScript on the client side for webpage behavior, often incorporating third-party libraries. All major web browsers have a dedicated JavaScript engine to execute the code on users' devices.
Linux	Linux is an open-source operating system that is popular in server environments and for development work. It requires specialized skills in installation, configuration, and maintenance, as well as proficiency in the use of command-line interfaces. Linux skills are highly valued in IT fields but may require additional training or certification for professionals who do not have prior experience with the operating system.

Apps & Tools Mentioned

Apps & Tools Title	Apps & Tools Description
Microsoft Azure	Microsoft Azure is a cloud computing platform that offers a wide range of services and tools to help organizations build, deploy, and manage their applications and infrastructure in the cloud. It provides services like virtual machines, storage, databases, analytics, networking, and more. Azure also supports a variety of programming languages, frameworks, and devices, making it a flexible and scalable option for businesses of all sizes. Specializing in Microsoft Azure involves learning the platform's key features and tools, including Azure Active Directory, Azure DevOps, Azure Functions, and Azure Kubernetes Service.
Microsoft CRM	Microsoft Dynamics CRM is a customer relationship management software package developed by Microsoft.
Microsoft Designer	Microsoft Design is a specialized skill related to creating software user interfaces and experiences using Microsoft's design software tools like Sketch, Figma, and Adobe XD. It involves creating intuitive, user-centered designs for desktop, web, and mobile applications that are visually appealing and easy to navigate. This skill is highly sought after by technology and design companies that specialize in creating software products.
Microsoft Excel	Microsoft Excel is a spreadsheet program developed and distributed by Microsoft Corporation. It is used to store, organize and analyze numerical data, and perform calculations such as financial modeling, statistical analysis, and budgeting. Excel also includes features like charts, tables, and pivot tables that allow users to visualize and understand data. It is a powerful tool widely used in business, finance, accounting, and other fields that deal with numbers and data.
Microsoft Office	Microsoft Office is a suite of software applications developed by Microsoft for use in both personal and professional settings. The suite includes programs such as Word, Excel, PowerPoint, and Outlook, which are commonly used for tasks such as document creation, data analysis, presentation design, and email communication. Microsoft Office is considered a common skill in the workplace, with proficiency in its various programs often required for many different types of jobs.
Microsoft PowerPoint	Microsoft PowerPoint is a presentation program, created by Robert Gaskins and Dennis Austin at a software company named Forethought, Inc. It was released on April 20, 1987, initially for Macintosh computers only.

Apps & Tools Mentioned

Apps & Tools Title	Apps & Tools Description
Microsoft Project	Microsoft Project is a project management software application developed by Microsoft. It allows project managers to plan, track, and manage projects of various sizes and complexities. The software includes features for scheduling, resource management, budgeting, task assignment, communication, and reporting. It is widely used in industries such as construction, engineering, manufacturing, IT, and business consulting. Microsoft Project is a specialized skill that requires knowledge and expertise in project management principles and practices, as well as proficiency in using the application.
Microsoft Teams	Microsoft Teams is a collaboration platform that enables teams to work together seamlessly, offering features such as chat, audio and video calls, file sharing, and integrations with other Microsoft applications like Outlook, OneDrive, and SharePoint. It also provides a virtual environment for holding remote meetings, conducting webinars and training, and has advanced security and compliance capabilities.
Microsoft Windows	Microsoft Windows is an operating system developed by Microsoft Corporation for personal computers. It provides a graphical user interface, file management, and communication capabilities. It has been one of the most widely used operating systems worldwide and offers a range of features and applications for an efficient and user-friendly computing experience.
Microsoft Word	Microsoft Word is a word processing software developed by Microsoft. It was first released on October 25, 1983, under the name Multi-Tool Word for Xenix systems.
Performance Analytics	Performance Analytics is a software tool that helps businesses measure and analyze data to improve their operations and decision-making. It allows users to visualize and track key performance indicators KPIs in real-time, compare data across time periods and departments, identify trends and outliers, and generate reports for stakeholders. This specialized skill requires knowledge of data analysis, statistics, and software tools such as Excel, Tableau, Power BI, or Google Analytics. Effective use of Performance Analytics can lead to better operational efficiency, increased revenue, cost savings, and improved customer satisfaction.
PracticeWorks	PracticeWorks is a specialized dental practice management software that automates several dental office tasks such as patient scheduling, billing, electronic claims, and dental charting. It enables dental practices to streamline their management, improve patient care, and enhance productivity. This software also helps dentists keep track of their patient's appointments, treatment history, and overall treatment plans, which leads to better patient outcomes.

Apps & Tools Mentioned

Apps & Tools Title	Apps & Tools Description
<i>Prevx</i>	Prevx is a security software that specializes in detecting and removing malware and other malicious software from a computer. It uses advanced technology to identify and eliminate threats before they can cause damage to the system. Additionally, Prevx provides real-time protection to prevent future infections and keeps the computer safe from online threats.
<i>Programming Environments</i>	Programming Environments refer to the integrated software tools that developers use to create, edit, and manage codes for developing software applications. These environments include code editors, compilers, interpreters, debuggers, and other tools that help programmers to create and test software efficiently. A programming environment must be reliable, easy to use, and adaptable to different programming languages to help developers to create effective and efficient software solutions.
<i>Project Management Information Systems</i>	Project Management Information Systems PMIS refer to the software applications and tools used by project managers to plan, organize, and manage projects from initiation to closure. PMIS provides project managers with insights on project progress, resource utilization, and risk assessments, helping them make informed decisions and ensure project success. This specialized skill requires knowledge of project management methodologies and experience with various PMIS tools such as Microsoft Project, Asana, Trello, and others.
<i>Project Management Software</i>	Project Management Software refers to any software tool that assists project managers in planning, tracking, and executing projects. It helps streamline project management processes, such as scheduling, budgeting, resource allocation, and team communication, to complete tasks within the project timeline and budget. Some popular project management software includes Asana, Trello, Microsoft Project, and Slack. Being proficient in project management software is a specialized skill that can increase productivity, enhance collaboration, and improve project outcomes.
<i>Python</i>	Python is a high-level, interpreted, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation.

Apps & Tools Mentioned

Apps & Tools Title	Apps & Tools Description
Reporting Tools	Reporting Tools is a specialized skill related to software that involves the ability to use various tools and techniques to generate reports and analyze data. This skill involves understanding data structures and formats, designing and creating reports, and interpreting data to provide insights and recommendations. Common tools used in reporting include Microsoft Excel, SQL, Oracle, Tableau, and Power BI, among many others. Effective reporting requires a strong attention to detail, exceptional organization and communication skills, and the ability to work with and interpret complex data.
SAP HR	SAP HR Human Resource management also known as Human Capital Management HCM and it is important module in ERP SAP systems that supports all the process of human resource management in the enterprise.
SAP Warehouse Management	SAP Warehouse Management SAP WM is an integrated software solution designed to manage and optimize warehouse operations, inventory management, and material movements in a company's supply chain. It provides real-time visibility of inventory levels, location tracking, and improves accuracy of stock movements based on best practices. SAP WM offers a range of features including managing goods receipt and issue processes, putaway and picking strategies, storage bin replenishment and optimization, cycle counting, and inventory management reporting. A specialized skill in SAP WM could involve configuring, customizing, implementing or providing support services for the software.
SAS	SAS previously "Statistical Analysis System" is a statistical software suite developed by SAS Institute for data management, advanced analytics, multivariate analysis, business intelligence, criminal investigation, and predictive analytics.
SQL	SQL is a domain-specific language used in programming and designed for managing data held in a relational database management system, or for stream processing in a relational data stream management system.
STATISTICA	STATISTICA is a statistical software package designed to perform data analysis, data management, data visualization and modeling, and to create reports and presentations. It is commonly used in research, academic and industrial organizations to analyze complex data sets and generate insights for decision making. The software offers a wide range of statistical calculations, tests, and models, allowing users to customize their analysis and generate accurate and reliable results. It is a powerful tool for statistical analysis that requires specialized knowledge and training to use effectively.

Apps & Tools Mentioned

Apps & Tools Title	Apps & Tools Description
ServiceNow	ServiceNow is a cloud-based enterprise software platform that provides a range of ITSM Information Technology Service Management solutions, as well as digital workflows for IT operations, customer service, HR, and security operations. It helps organizations automate their processes, improve efficiency, and enhance collaboration among teams. ServiceNow is a highly sought-after skill for IT professionals and can lead to a variety of job opportunities in different industries.
Simpletest	Simpletest is a software testing framework that focuses on writing and executing automated tests for web applications. It is designed to be simple and easy to use, making it accessible for developers of any skill level. Simpletest provides a variety of features such as test fixtures, assertions, and mocking, that enable users to write comprehensive and efficient tests for their web applications.
Socialengine	Socialengine is a highly customizable white-label social networking software that allows businesses, organizations, and individuals to create their own social network websites. With its powerful features, users can create a unique online community tailored to their specific needs and preferences. Some key features of Socialengine include user profiles, network statistics, social networking tools, forums, blogs, photo and video galleries, and much more. It is a specialized skill to be able to effectively customize and manage Socialengine platforms.
Supervisord	Supervisord is a process control system used to monitor and manage processes on a Unix or Linux system. It provides a way to control and manage background processes and applications, ensuring they are started, stopped, and restarted as needed. Supervisord is often used in server environments where multiple applications need to run continuously, and uptime and stability are critical. It can be configured using a simple configuration file and has a web interface for managing and monitoring its processes.
Windows Search	Windows Search is a software program that allows users to quickly search for specific files and content on their Windows computer. It is designed to make it easier for users to find the information they need, by indexing files and documents on the computer and providing relevant search results in a matter of seconds. This specialized skill involves knowledge of the Windows operating system and the ability to use features such as search filters, indexing options, and advanced search techniques.
Wordfast	Wordfast is a computer-assisted translation CAT software tool used by professional translators that helps to improve productivity and efficiency in the translation process by automating certain tasks such as segmenting text, creating translation memories and terminology databases, and aligning documents for bilingual comparison. This specialized skill requires knowledge of how to use the software effectively, as well as expertise in translation and language.

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